University of Wollongong

Bachelor of Computer Science

CSIT321 - Final Year Project

A Mobile Application for Wise Workout

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A Mobile Application for Wise Workout

Revision History

Revision Number	Revision Date	Revised by	Remarks

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1. Introduction

1.1. Overview

Since the 1970s, wearable technology for health and fitness has undergone a remarkable evolution, moving from simple, standalone devices to advanced ecosystems that seamlessly integrate hardware, software, and data analytics. This shift has transformed consumer health behaviors, the fitness industry, and healthcare delivery, with the global wearable market expected to reach $25.9 billion by 2033, driven by a compound annual growth rate (CAGR) of 18% (Revankar, 2024).

This presents a strategic opportunity to leverage the market through a service platform for smart fitness, emphasizing a cohesive ecosystem approach. The WiseWorkout App aims to create a mobile platform that unifies smart devices and wearables into a dynamic ecosystem, delivering real-time performance analytics, intelligent health and wellness insights, and an engaging interface for users to challenge their psychological limits, enabling them to achieve attainable goals without requiring a physical coach.

1.2. Project Scope

The WiseWorkout App aims to be a central platform that aggregates exercise data from smartphone sensors and wearable devices. Its core objective is to provide insightful analytics and smart recommendations, enabling fitness enthusiasts to make data-driven decisions for their wellness journey.

Key features include the “short- and long-term exercise effect estimate,” delivering in-depth analytics such as calories burned, average and peak heart rate, and progress tracking across flexible timeframes (daily, weekly, monthly, or custom). Powered by AI, the app offers tailored fitness advice based on user goals, health data, and activity history. It also supports customizable fitness plans, allowing users to adjust parameters like plan length or daily weight loss goals. Smart notifications will prompt users to stay active or rest, promoting balanced routines.

1.3. Project Description

The app encourages social interaction by allowing users to share workout data and competition outcomes on platforms like Facebook, Instagram, Twitter, and TikTok. Users can launch challenges with others, with shareable rankings to boost motivation. Designed with a responsive, intuitive interface, the app ensures smooth operation on Android and iOS devices. The project plans to pursue app store distribution, leveraging a scalable business model to tap into the expanding wearable technology market.

1.4. Business Model

Key Partners

The key partners in our hospital model are:

1. Trainers
2. Subject matter experts

Key Activities

Our key activities in the development of Wise focus on integrating smart features while keeping the user interface clean. We aim to deliver a user-friendly experience that allows seamless fitness tracking and personalized goal management. Collaboration with fitness trainers and subject matter experts ensures that the advice, fitness plans, and analytics provided in the app are accurate, safe, and effective. This partnership helps us maintain the reliability and credibility of our service while constantly improving the content and features to meet user needs.

Key Resources

The following key resources that our business model has adopted are:

- Development Team
- Wearable & Sensor Integration
- Health & Fitness Data
- Fitness Trainers & Subject Matter Experts
- Databases
- AI & Analytics Engine

Value Propositions

Wise Workout App is designed to be platform-agnostic and inclusive, offering a comprehensive suite of data-driven insights to help users make informed health and fitness decisions—minimizing the need for a personal trainer. Our core philosophy centers on empowerment through personalized experiences and customizable plans, while also promoting community engagement and social interaction to keep users motivated and connected.

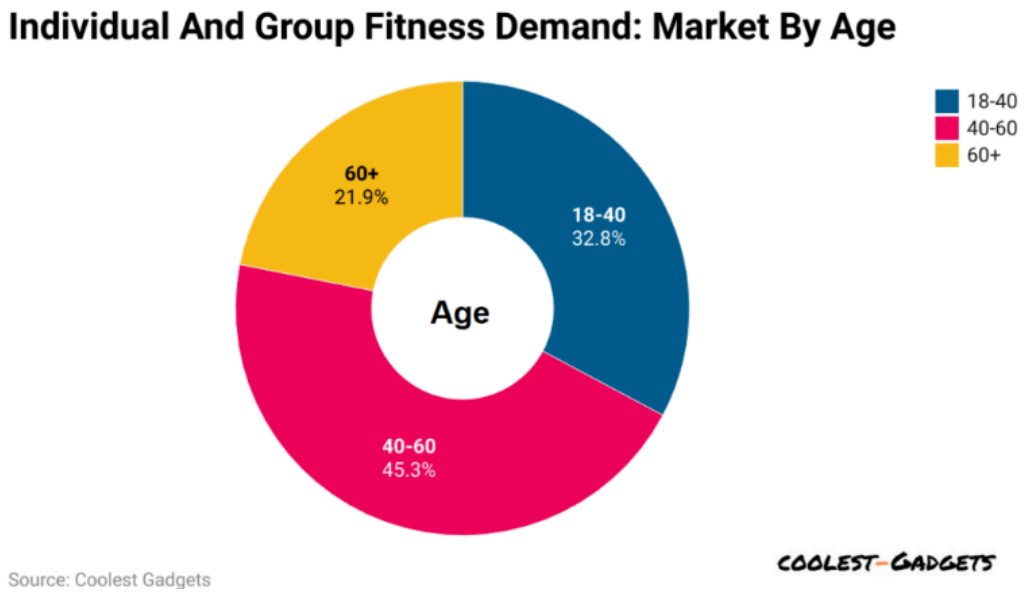


Table 1 Source: https://www.coolest-gadgets.com/fitness-industry-statistics/ (2024)

Individual User (Consumers)

The WiseWorkout App targets the 18–40 age range (Gen Z and Millennials), who form 32.8% of the fitness market, as its primary customer base, with a strategy focused on early acquisition to foster long-term brand loyalty. Additionally, the app aims to attract and retain users aged 40 and beyond (Gen X and older), leveraging its scalable features to grow these segments with the brand over time, capitalizing on the significant market potential of Gen X, who represent nearly half the population.

Also, positioned as an app for casual users striving for self-improvement, WiseWorkout caters to health-conscious individuals, from novice gym-goers seeking guidance to busy professionals needing convenient home workouts.

Business Users

- Fitness Centers/ Gym
- Personal Trainers/ Fitness Coaches
- Wearable Device Companies
- Health insurance companies
- Advertisers/ Sponsors

Channels

- Mobile App stores (iOS, Android)
- Social media (Instagram, Tiktok fitness influencers)
- Fitness blog, YouTube
- Partnerships with gyms, training, supplement brands

Revenue Stream

The WiseWorkoutApp will adopt a hybrid business model to capitalize on both the growing consumer demand for personalized health solutions and the expansion of the wearable tech ecosystem. The model integrates multiple revenue streams to ensure scalability, sustainability, and broad user adoption.

Freemium Access with Tiered Subscriptions

Users can access a core set of features for free, including basic activity tracking, short-term analytics, and integration with common wearable devices. Premium users unlock advanced features such as:

Premium Access

Subscriptions are available monthly, quarterly, or annually, priced competitively to appeal to both casual users and dedicated fitness enthusiasts.

In-App Purchase

The app offers one-time purchases of:

- Specialized workout plans (e.g., marathon training, postpartum recovery)
- Nutritional guides and meal plan templates
- Avatar upgrades and achievement badges for gamification

Cost Structure

A general overview of the spending and expenditure for this project are broken down as followed:

- Development Cost
    - Consist of Research and Developement
    - Software Development (IDEs, SDKs, Emulator, cloudtools (e.g., firebase, github taiga)
    - Cross-platform framework (flutter)
    - APIs and SDKs (integration with sensors)
    - Backend Services (Cloud based)
    - Wearable Devices Testing (Borrowed, purchased)
    - Mobile Devices for testing etc..

- Operation Cost
    - App Store Developer Acc (iOS, android)
    - Marketing and Promo (social media campaign, promo)
    - Domain and Web Hosting (for website)
    - Project Management Tools (Ms Project)
    - Project presentation tools (Video editing tools)

- Human Resources Cost
    - Student Dev Time
    - Supervisor Consultation

- Contingency and Miscellaneous
    - MISC Expenses
    - Contingency Buffer (10%)

Customer Segments

Our customer segments will mainly be the people who will be using our app. This comprises all users and business users of different categories.

- Business User: To post any of their services such as pt trainers or tips easily
- Freemium User: Offer core features for them so they can track their activity easily.
- Premium User: Can access what normal users can do but also more in depth analytics.

2. Research Finding Analysis

2.1 Fitness Application in the market

Table 2 image taken from https://www.businessofapps.com/news/health-and-fitness-app-market-map-launched/

2.2 Research summary

Fitbod

Smart Features: Generates custom strength workouts based on user history, available equipment, and fitness goals.

AI/ML: Employs machine learning to predict optimal recovery periods and progression paths.

Tech Stack: Native iOS application with on-device machine learning; integrates with Apple HealthKit.

Monetization Model: Subscription-based at $12.99/month or $79.99/year.

Freeletics

Smart Features: Provides adaptive coaching that adjusts routines based on user performance and feedback.

AI/ML: Utilizes AI to adapt workout intensity through performance feedback loops.

Tech Stack: Flutter front-end with a custom AI coaching engine.

Monetization Model: Freemium model with a subscription-based Coach at approximately $34.99 per quarter.

Apple Fitness+

Smart Features: Offers a variety of workouts across different categories, with real-time metrics when paired with Apple Watch.

AI/ML: Leverages on-device machine learning through Core ML for personalized recommendations.

Tech Stack: Built into the Apple ecosystem, integrating seamlessly with iOS, watchOS, and Apple HealthKit.

Monetization Model: Subscription-based at $9.99/month or $79.99/year; also available through Apple One bundles.

Strava

Smart Features: Combines fitness tracking with social engagement, including leaderboards and community challenges.

AI/ML: Introduced "Athlete Intelligence" for personalized workout feedback and uses machine learning to flag suspicious activities, ensuring leaderboard integrity.

Tech Stack: Utilizes a combination of mobile and web platforms; integrates with various wearables and third-party apps.

Monetization Model: Freemium model with a premium subscription offering advanced features.
Suffescom Solutions

Nike Run Club (NRC)

Smart Features: Provides guided runs, personalized coaching plans, and real-time audio feedback.

AI/ML: Incorporates AI to offer personalized coaching experiences and adaptive training plans.

Tech Stack: Native mobile applications with integration into the Nike ecosystem and Apple HealthKit.

Monetization Model: Free to use, supporting Nike's broader brand engagement strategy.
TechRadar

Runkeeper

Smart Features: Tracks various activities, offers personalized coaching, and allows for goal setting and progress tracking.

AI/ML: Employs AI for adaptive coaching and performance analysis.

Tech Stack: Native mobile applications; integrates with wearables and health platforms.

Monetization Model: Freemium model with a premium subscription offering additional features.

2.2 Product comparison

Product Comparison

Function/ Features	Fitbod	Freeletics	Apple Fitness	Strava	Nike Run Club (NRC)	Runkeeper	Our Product
Activity Tracking (Steps, Workouts)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Health Metrics (Heart Rate, Calories)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Exercise Effect Estimation	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Personalized Fitness Plans/ Advice	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Smart Reminders & Notification	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Social Features (Challenge and LeaderBoards)	No	Yes	Yes	Yes	Yes	Yes	Yes
Social Media Sharing	No	Yes	Yes	Yes	Yes	Yes	Yes
Wearables Integration	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nutrition Tracking	No	No	No	No	No	Yes	Yes
AI-Driven Insights/Adaptation	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Gamification (Badges, Rewards)	No	Yes	Yes	Yes	Yes	Yes	No
Data Visualization (Trends, Reports)	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cross-Platform Support (iOS & Android)	Yes	Yes	No	Yes	Yes	Yes	Yes

Add on: Optional/ Differentiating Factor							
Function/ Features	**Fitbod**	**Freeletics**	**Apple Fitness**	**Strava**	**Nike Run Club (NRC)**	**Runkeeper**	**Our Product**
Team mode							Yes
Habit Tracker							Yes
Carbon footprint tracker							Yes
SOS Safety Alert							Yes
Voice Assistance Support							Good to have
Songs Recommendation to pair with workout (Spotify/ YT/ Apple)							Good to have
Voice-Activated AI Coach							Good to have

3. Project Motivation

With the rise of fitness awareness and the increasing demand for healthier lifestyles, many individuals are seeking convenient and personalized tools to help them stay active, track progress, and achieve fitness goals. However, many existing apps focus only on basic tracking, leaving gaps in smart analysis, customizable planning, and interactive features.

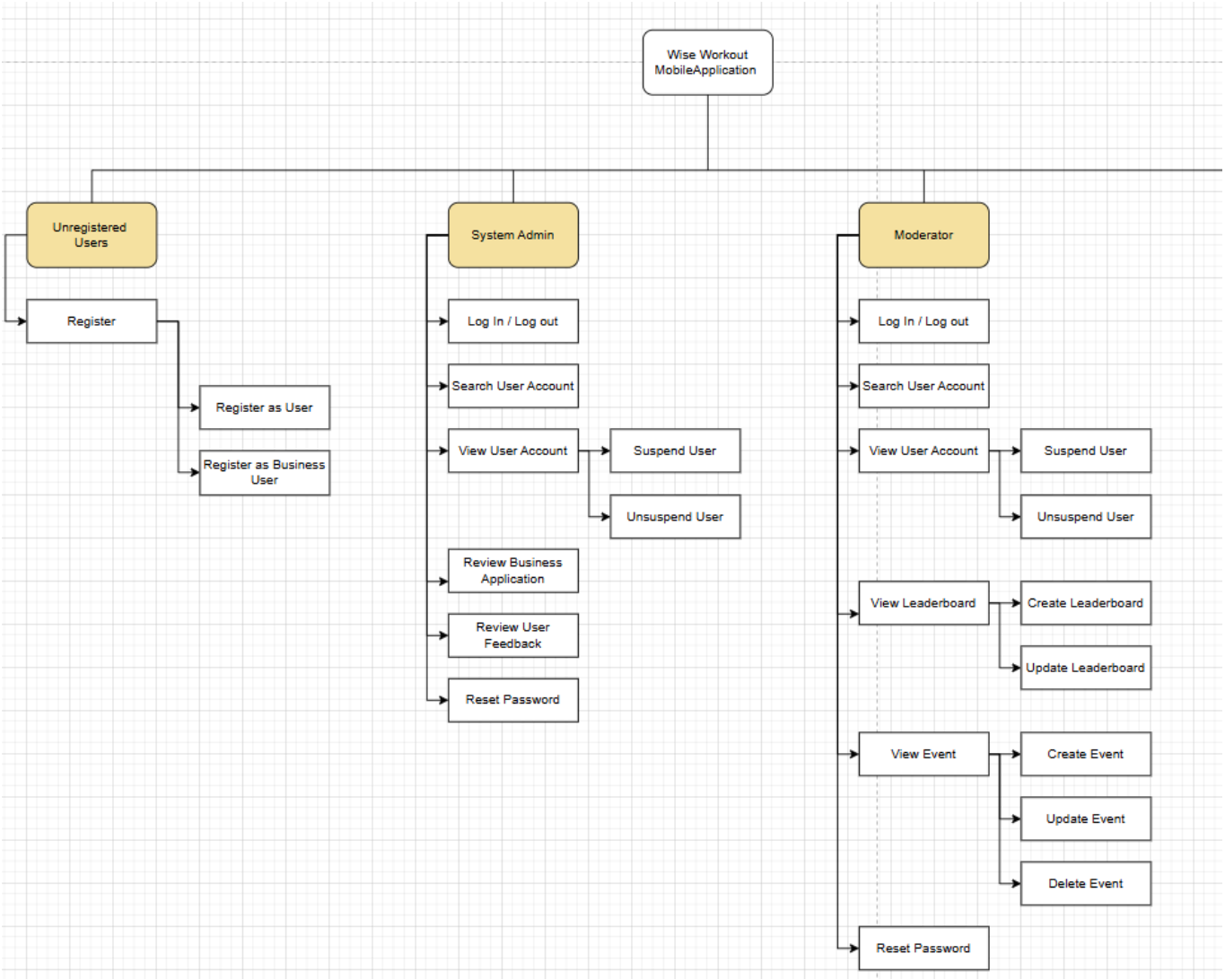
Additionally, people often struggle with staying motivated, especially when working out alone. There is also a growing interest in social fitness experiences and integrating fitness into everyday routines.

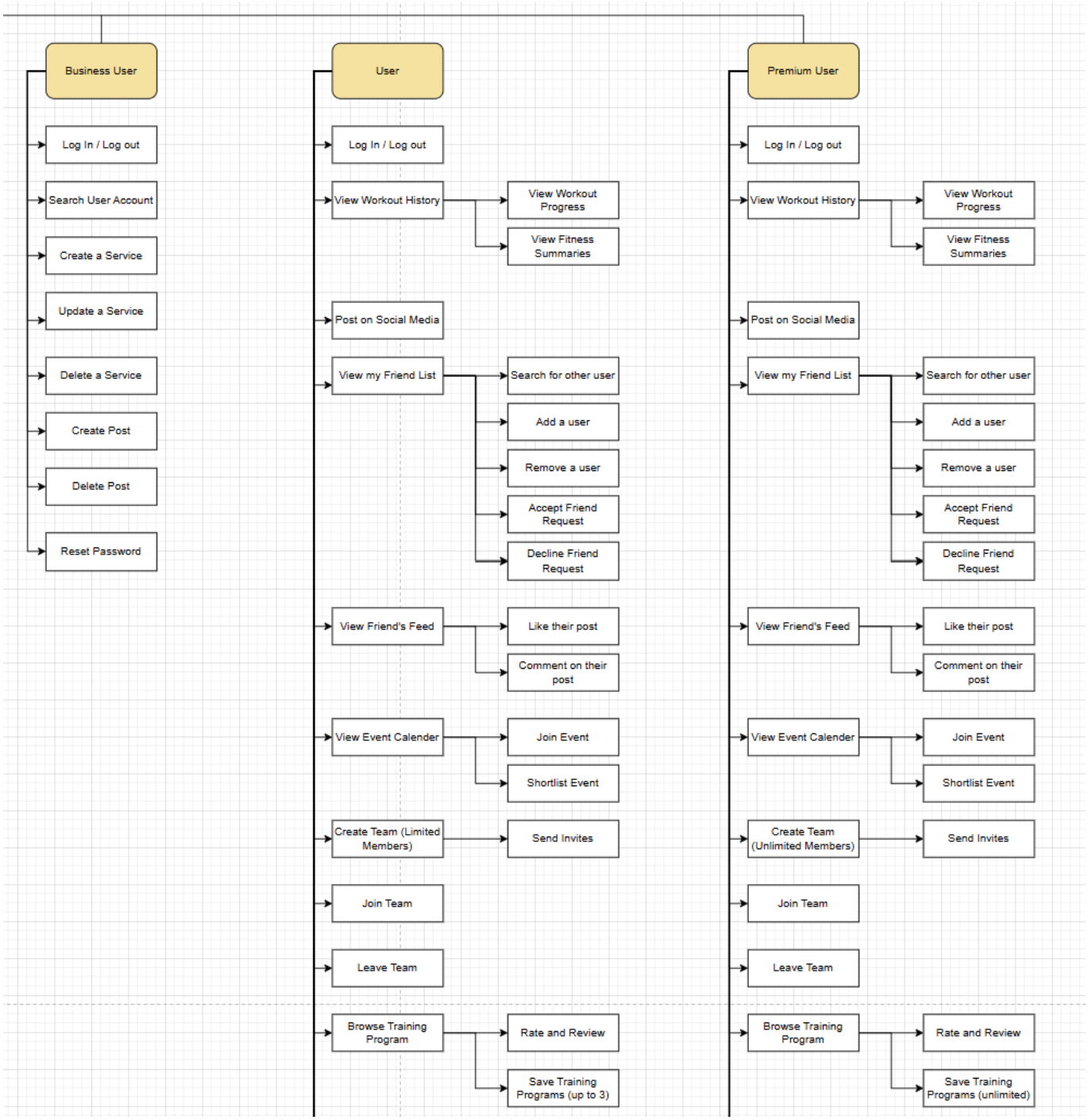
Wise App is motivated by the need to:

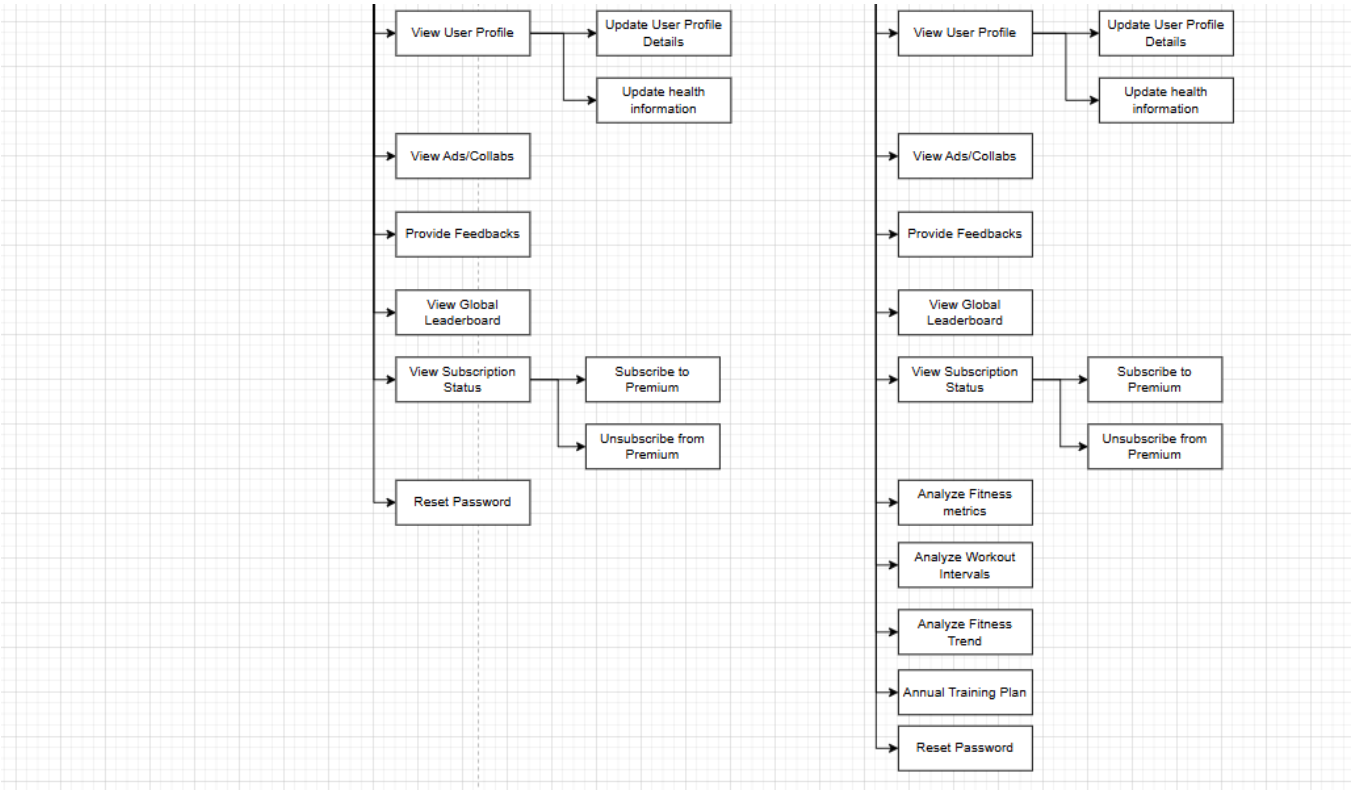
- Automate and simplify fitness tracking and planning using AI and sensor data.
- Offer a more personalized and adaptable fitness experience.
- Provide on-the-go access to health metrics, advice, and reminders.
- Reduce the dependency on manual planning or gym visits.
- Encourage consistency through social interaction, challenges, and friendly competition.
- Empower users with actionable insights and habit tracking to build long-term healthy routines.

4. Functional Requirements

4.1 Functional Hierarchy

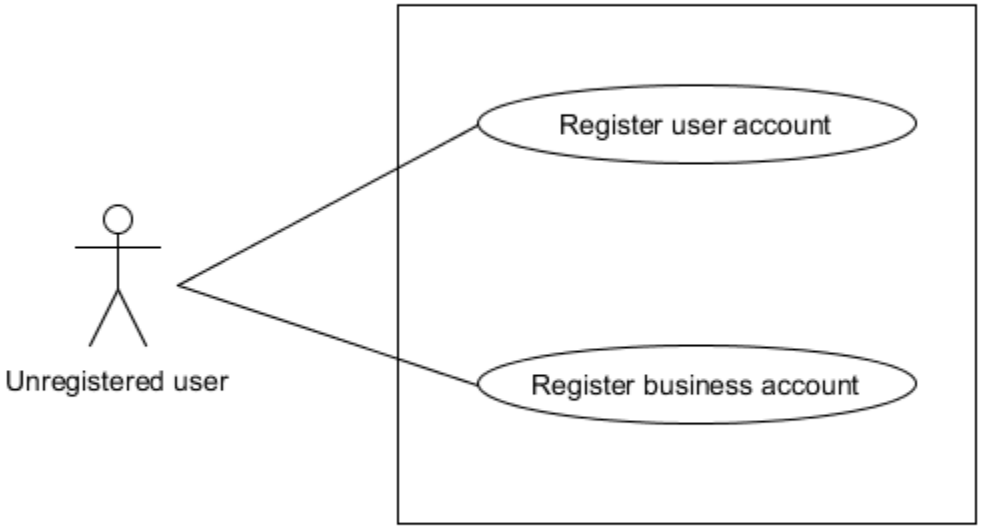



Business User
Log In / Log out
Search User Account
Create a Service
Update a Service
Delete a Service
Create Post
Delete Post
Reset Password
User
Log In / Log out
View Workout History
View Workout Progress
View Fitness Summaries
Post on Social Media
View my Friend List
Search for other user
Add a user
Remove a user
Accept Friend Request
Decline Friend Request
View Friend's Feed
Like their post
Comment on their post
View Event Calender
Join Event
Shortlist Event
Create Team (Limited Members)
Send Invites
Join Team
Leave Team
Browse Training Program
Rate and Review
Save Training Programs (up to 3)
Premium User
Log In / Log out
View Workout History
View Workout Progress
View Fitness Summaries
Post on Social Media
View my Friend List
Search for other user
Add a user
Remove a user
Accept Friend Request
Decline Friend Request
View Friend's Feed
Like their post
Comment on their post
View Event Calender
Join Event
Shortlist Event
Create Team (Unlimited Members)
Send Invites
Join Team
Leave Team
Browse Training Program
Rate and Review
Save Training Programs (unlimited)

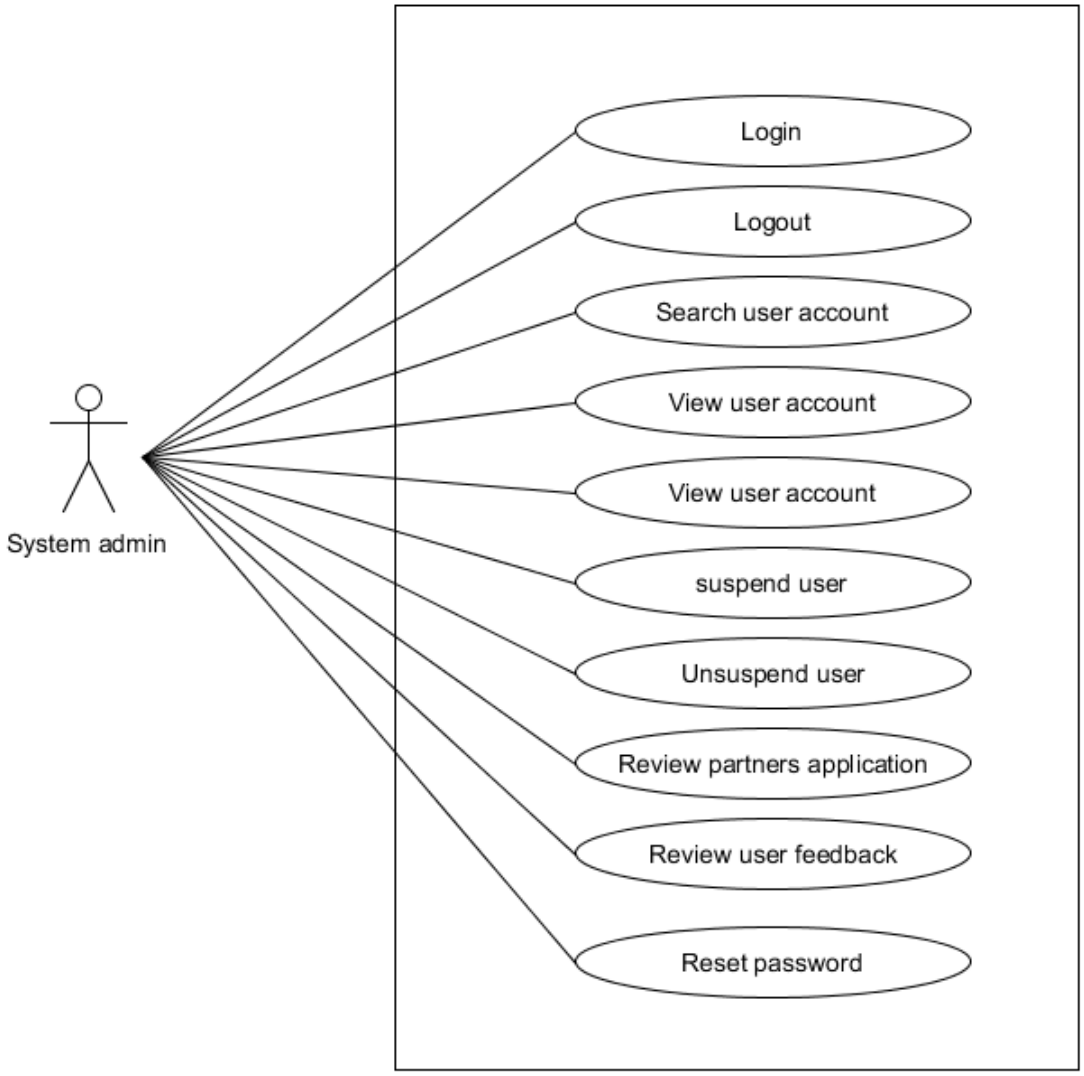

View User Profile
Update User Profile Details
Update health information
View Ads/Collabs
Provide Feedbacks
View Global Leaderboard
View Subscription Status
Subscribe to Premium
Unsubscribe from Premium
Reset Password
View User Profile
Update User Profile Details
Update health information
View Ads/Collabs
Provide Feedbacks
View Global Leaderboard
View Subscription Status
Subscribe to Premium
Unsubscribe from Premium
Analyze Fitness metrics
Analyze Workout Intervals
Analyze Fitness Trend
Annual Training Plan
Reset Password

4.2 Preliminary Use Case Diagram

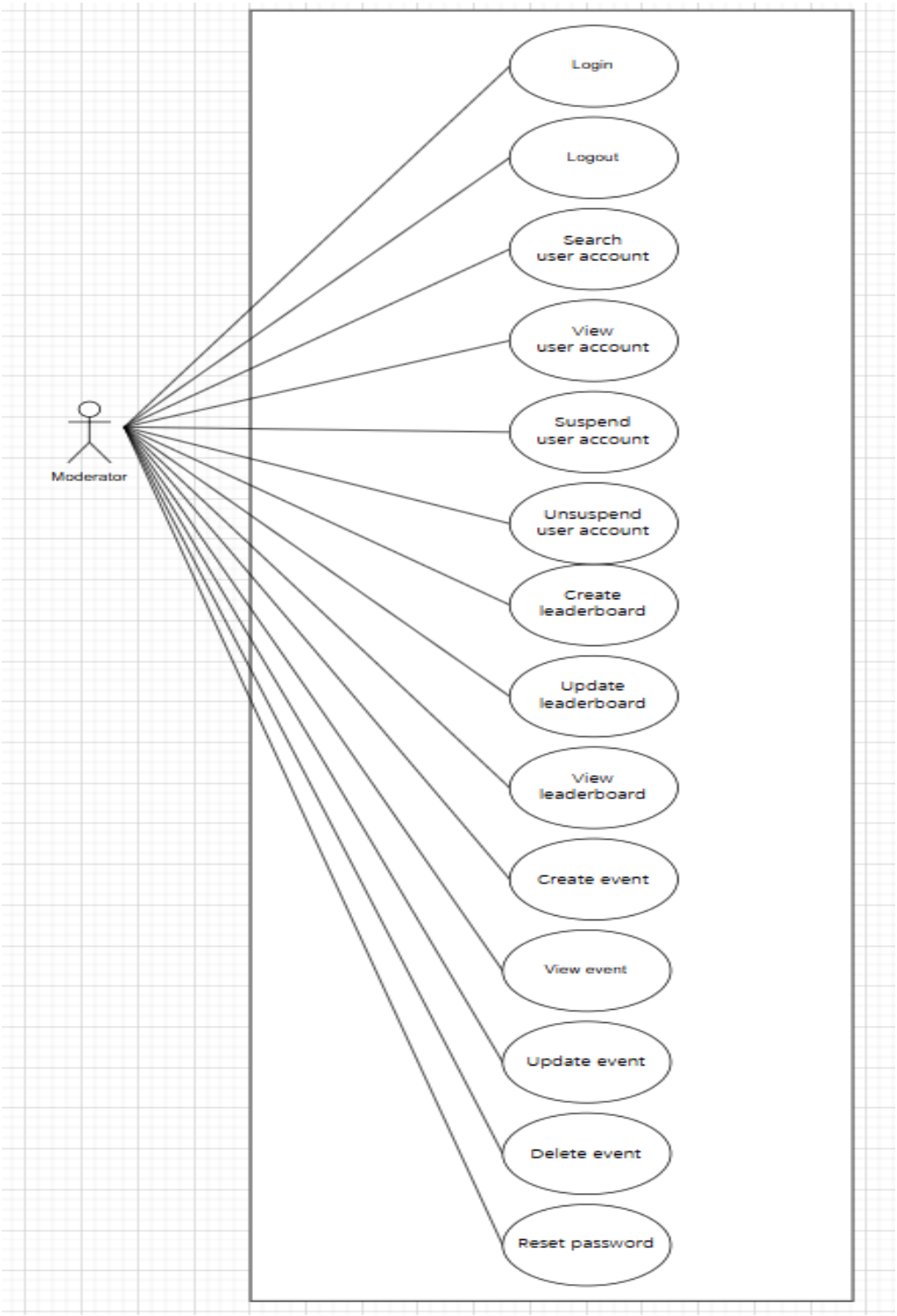
Unregister User



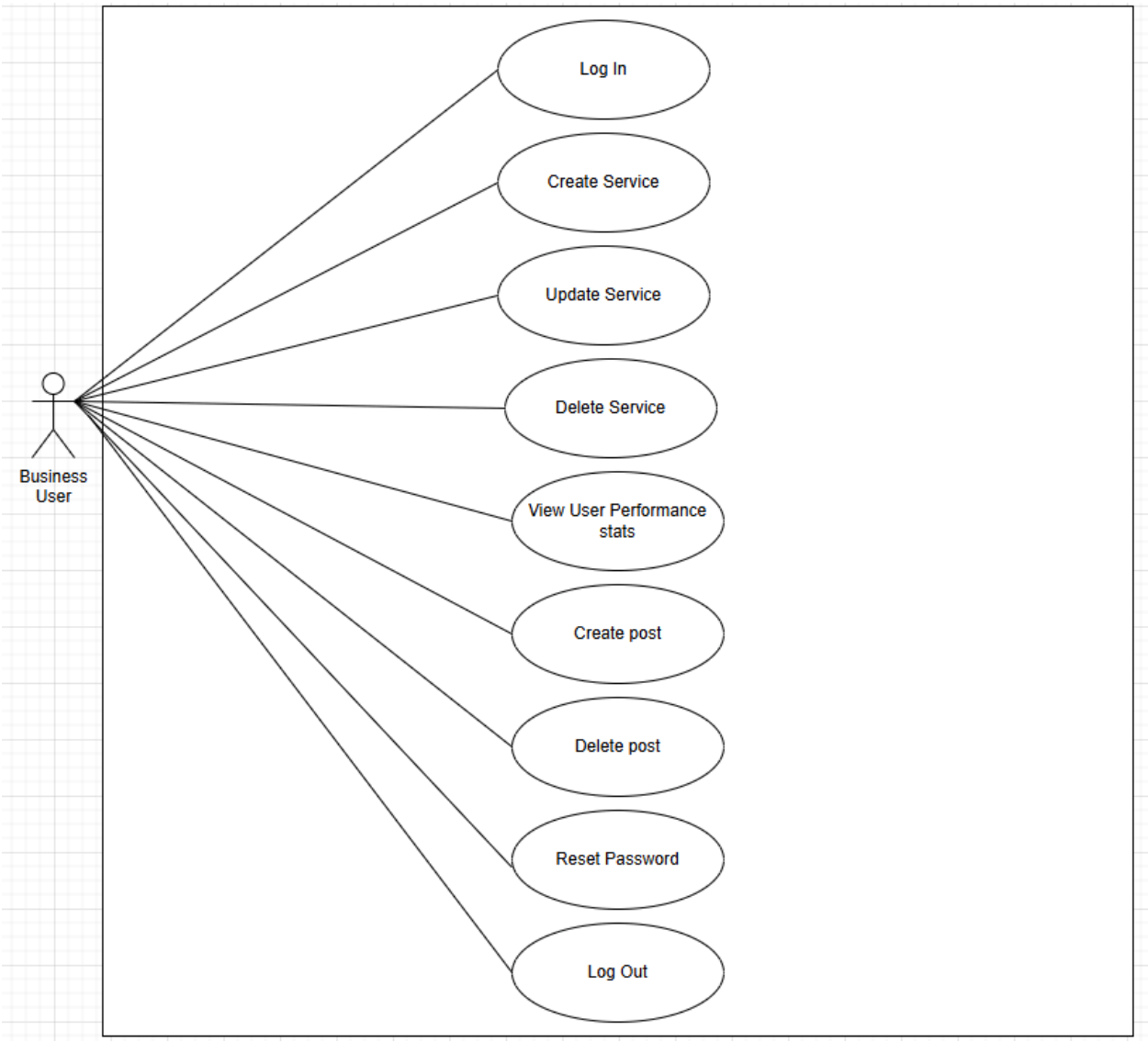
System Admin



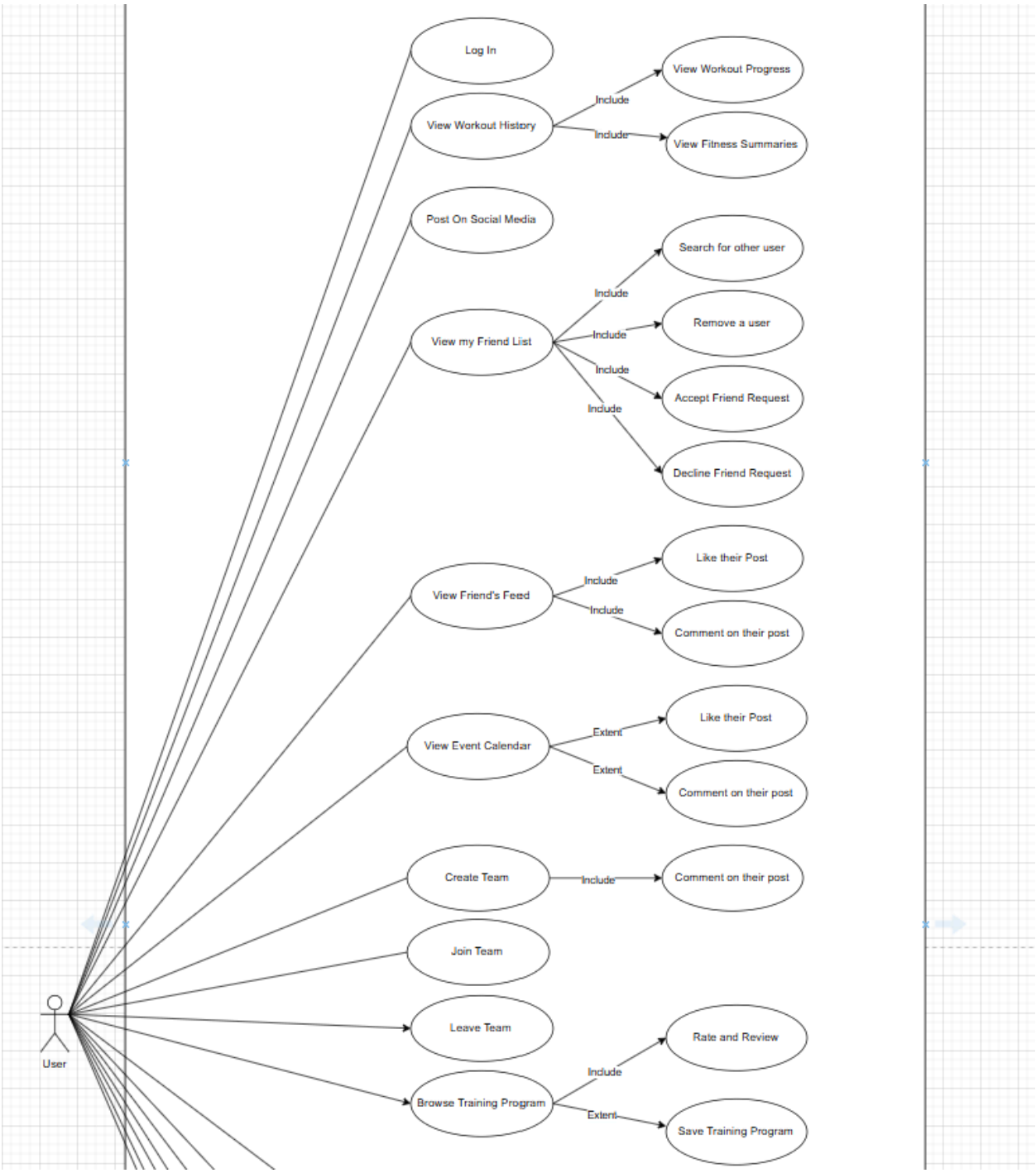
Moderator

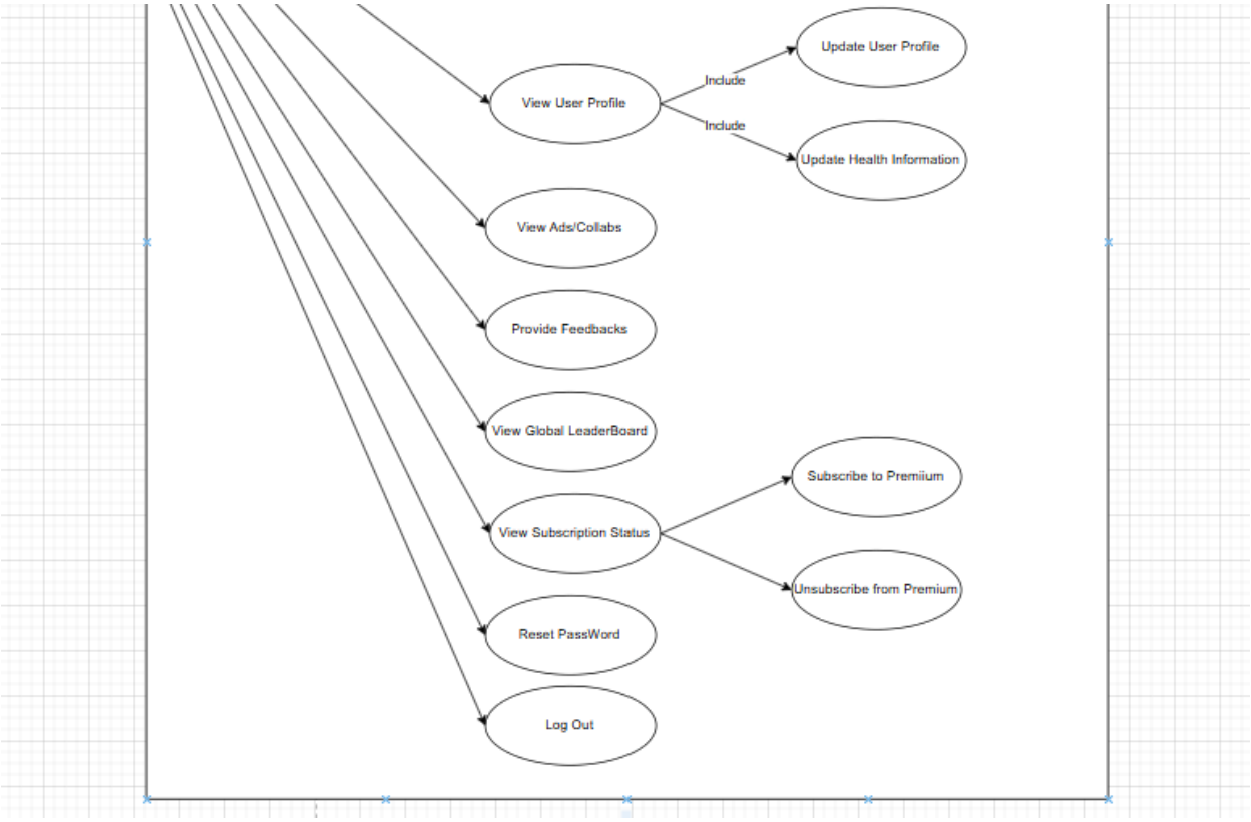

Moderator
Login
Logout
Search user account
View user account
Suspend user account
Unsuspend user account
Create leaderboard
Update leaderboard
View leaderboard
Create event
View event
Update event
Delete event
Reset password

Business User

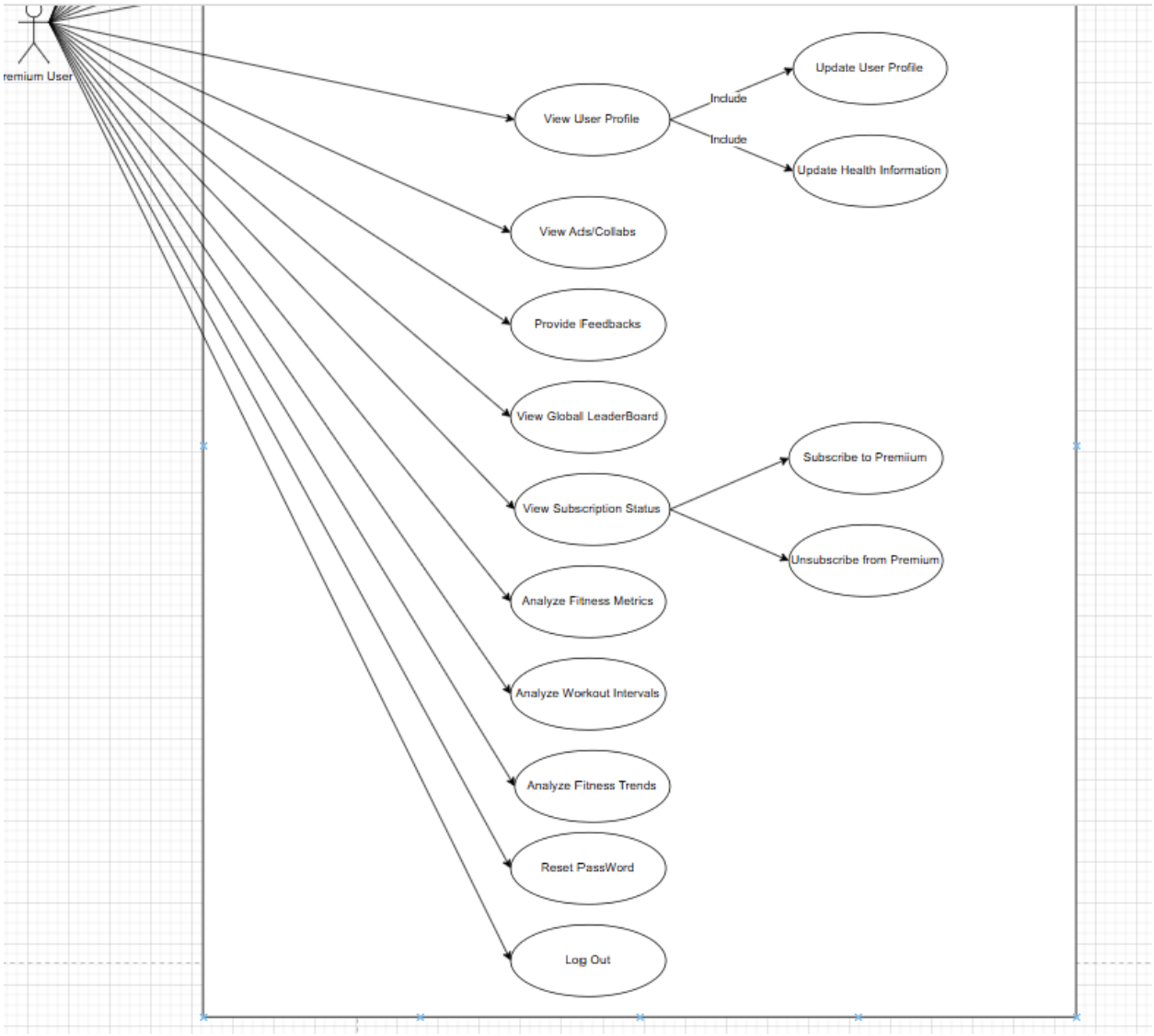


User


Log In
View Workout History
Include
View Workout Progress
Include
View Fitness Summaries
Post On Social Media
View my Friend List
Include
Search for other user
Include
Remove a user
Include
Accept Friend Request
Include
Decline Friend Request
View Friend's Feed
Include
Like their Post
Include
Comment on their post
View Event Calendar
Extent
Like their Post
Extent
Comment on their post
Create Team
Include
Comment on their post
Join Team
Leave Team
Browse Training Program
Include
Rate and Review
Extent
Save Training Program
User


View User Profile
Include
Update User Profile
Include
Update Health Information
View Ads/Collabs
Provide Feedbacks
View Global LeaderBoard
View Subscription Status
Subscribe to Premiium
Unsubscribe from Premium
Reset PassWord
Log Out

Premium User


remium User
View User Profile
Include
Include
Update User Profile
Update Health Information
View Ads/Collabs
Provide Feedbacks
View Global LeaderBoard
View Subscription Status
Subscribe to Premiium
Unsubscribe from Premium
Analyze Fitness Metrics
Analyze Workout Intervals
Analyze Fitness Trends
Reset PassWord
Log Out

4.3 Preliminary Use Case Description

4.3.1 Unregistered User Use Case Description

Name: Register User Account	**ID: #01**
Stakeholders and goals: Unregistered User – To create a personal account for accessing app features.	
Description: An unregistered user fills in a registration form to create a new personal user account.	
Actors: Unregistered User / Guest	
Trigger: When the user clicks the “Register” or “Sign Up” button.	
Pre-condition: The user is not already registered and has internet access.	
Main flow: 1. User launches the app or website. 2. Click the “Sign Up” button. 3. Select the “User Account” option. 4. System displays a registration form. 5. User fills in details (name, email, password, etc.). 6. User agrees to terms and conditions. 7. User clicks “Register”. 8. System validates the input. 9. System creates the account and logs the user in.	

10. User is redirected to the dashboard.
11. End.

Sub- flow: None.

Alternative/Exceptional flow:

5a. Missing fields or invalid input

i. System displays error message.

ii. Return to Step 5.

Name: Register Business User	**ID: #02**

Stakeholders and goals: Guest – To register a business account for offering services or partnerships.

Description: A guest fills in the business registration form to apply as a partner/business user.

Actors: Unregistered User / Guest

Trigger: When the user clicks the “Register as Business” button.

Pre-condition: The user is not already registered as a business and has access to required documentation.

Main flow:

1. User visits the registration section.

2. Click the “Register as Business” button.
3. System displays a business registration form.
4. User enters business name, contact details, purpose, uploads necessary documents.
5. User agrees to terms and conditions.
6. Click “Submit”.
7. System stores applications.
8. Displays confirmation that application is under review.
9. End.

Sub- flow: None.

Alternative/Exceptional flow:

4a. Missing mandatory fields or invalid file upload

i. System shows error message.

Name: Login	**ID: #03**
Stakeholders and goals: System Admin – To access the system securely	
Description: The System Admin enters credentials and clicks the “Login” button to access the admin dashboard	

Actors: System Admin
Trigger: When the admin enters login credentials and clicks "Login"
Pre-condition: The System Admin account exists and is active
Main flow: 1. Admin opens the app. 2. System shows a login screen. 3. Admin enters username and password. 4. Admin clicks "Login" 5. System validates credentials. 6. Admin is redirected to the dashboard. 7. End.
Sub- flow: None.
Alternative/Exceptional flow: 5a. Blank input or incorrect credentials i. System shows an error message: “Invalid username or password.” ii. Return to Step 3.

Name: Logout	**ID: #04**
Stakeholders and goals: System Admin – To securely end the session and log out.	
Description: The System Admin logs out of the system to terminate the session securely.	
Actors: System Admin	
Trigger: When the admin clicks the "LogOut" button.	
Pre-condition: The System Admin must be logged into the system.	
Main flow: 1. Admin clicks "Log Out". 2. System logs out the admin. 3. Admin is redirected to the login screen. 4. End.	
Sub- flow: None.	
Alternative/Exceptional flow: None	

Name: Search User account	**ID: #05**
Stakeholders and goals: System Admin – To search for specific user accounts in the system.	

Description: The System Admin searches for specific user accounts by various criteria.
Actors: System Admin
Trigger: When the admin enters search criteria and clicks "Search".
Pre-condition: The System Admin must be logged into the system.
Main flow: 1. Admin navigates to the search user screen. 2. System shows the search interface. 3. Admin enters search criteria (username, email, etc.). 4. Admin clicks "Search". 5. System displays matching user accounts. 6. End.
Sub- flow: None.
Alternative/Exceptional flow: 3a. No matching results i. System displays a message: "No users found with the given criteria." ii. Return to Step 3.

Name: View user account details	**ID: #06**
Stakeholders and goals: System Admin – To view detailed information about a user account.	
Description: The System Admin views detailed information of a specific user account to verify status and history.	
Actors: System Admin	
Trigger: When the admin selects a user account from the search results or user list.	
Pre-condition: The System Admin must be logged in and have access to the user account information.	
Main flow: 1. Admin selects a user account from the list. 2. System retrieves and displays the user's details (status, history, etc.). 3. Admin views the information. 4. End.	
Sub- flow: None.	
Alternative/Exceptional flow: None	

Name: Suspend User	**ID: #07**
Stakeholders and goals: System Admin – To suspend a user account to enforce community guidelines.	
Description: The System Admin suspends a user account that violates guidelines or policies.	
Actors: System Admin	
Trigger: When the admin selects the "Suspend" option for a user account.	
Pre-condition: The System Admin must be logged in and have the required permissions to suspend user accounts.	
Main flow: 1. Admin selects a user to suspend. 2. System prompts for confirmation. 3. Admin confirms the suspension. 4. System suspends the user account and sends a notification to the user. 5. End.	
Sub- flow: None.	
Alternative/Exceptional flow: 3a. Admin cancels suspension i. Admin decides not to suspend the user. ii. Return to Step 1.	

Name: Unsuspend User	**ID: #08**
Stakeholders and goals: System Admin – To unsuspend a user account after the issue has been resolved.	
Description: The System Admin unsuspends a previously suspended user account.	
Actors: System Admin	
Trigger: When the admin selects the "Unsuspend" option for a suspended user account.	
Pre-condition: The System Admin must be logged in and have the required permissions to unsuspend user accounts.	
Main flow: 1. Admin selects a suspended user account. 2. System prompts for confirmation to unsuspend. 3. Admin confirms the unsuspension. 4. System unsuspends the user account and sends a notification to the user. 5. End.	
Sub- flow: None.	
Alternative/Exceptional flow: None	

Name: Review Partner Applications	**ID: #09**
Stakeholders and goals: System Admin – To review partner applications for approval.	
Description: The System Admin reviews partner applications to determine whether they meet the criteria for approval.	
Actors: System Admin	
Trigger: When the admin navigates to the partner applications section and reviews the applications.	
Pre-condition: The System Admin must be logged in and have access to the partner application section.	
Main flow: 1. Admin navigates to the "Partner Applications" section. 2. System displays a list of partner applications. 3. Admin reviews each application. 4. Admin approves or rejects the application. 5. System updates the partner status and notifies the applicant. 6. End.	
Sub- flow: None.	

Alternative/Exceptional flow: None

Name: Review User Feedback	**ID: #10**
Stakeholders and goals: System Admin –To review user feedback and improve the app experience.	
Description: The System Admin reviews feedback submitted by users to identify areas of improvement for the app.	
Actors: System Admin	
Trigger: When the admin selects the "Review Feedback" option.	
Pre-condition: The System Admin must be logged in and have access to user feedback data.	
Main flow: 1. Admin navigates to the "User Feedback" section. 2. System displays all the user feedback submitted. 3. Admin reviews the feedback. 4. Admin analyzes the feedback and takes action to improve the app experience. 5. End.	
Sub- flow: None.	

Alternative/Exceptional flow: None

Name: Reset User Password	**ID: #11**
Stakeholders and goals: System Admin –To reset the password of a user who is locked out.	
Description: The System Admin resets a user's password to assist with account access.	
Actors: System Admin	
Trigger: When the admin selects the "Reset Password" option for a user.	
Pre-condition: The System Admin must be logged in and have the necessary permissions to reset user passwords.	
Main flow: 1. Admin selects a user whose password needs resetting. 2. System prompts the admin to reset the password. 3. Admin confirms the reset and enters a new password. 4. System updates the user's password and sends a notification to the user. 5. End.	
Sub- flow: None.	
Alternative/Exceptional flow: None	

4.3.3 Moderator Use Case Description

Name: Moderator - Log In	**ID: #12**
Stakeholders and goals: Moderator – To log in to the system	
Description: Moderator inputs their credentials and clicks the “Log In” button to access the system.	
Actors: Moderator	
Trigger: When the Moderator enters login details and clicks on the Log In button.	
Pre-condition: The Moderator account has been successfully created	
Main flow: 1.Moderator launches the app. 2.System displays fields for username and password. 3.Moderator inputs username and password. 4.Moderator clicks the “Log In” button. 5.System validates the username, password, and profile. 6.Moderator is redirected to the dashboard. 7.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 5.1a. Blank input i.System prompts “Please fill up username/password.” 5.1b. Repeat steps 3 and 4.	

5.2a. Invalid login i. System displays “Invalid username/password.” 5.2b. Repeat steps 3 and 4.

Name: Moderator - Log Out	**ID: #13**
Stakeholders and goals: Moderator – To log out from the system	
Description: Moderator clicks the “Log Out” button to end the session.	
Actors: Moderator	
Trigger: When the Moderator clicks the Log Out button	
Pre-condition: Moderator is logged in	
Main flow: 1.Moderator clicks the “Log Out” button. 2.System ends the session and redirects to the login page. 3.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 2a. Session timeout i. System automatically logs out the moderator and redirects to login.	

Name: Moderator - Search User Accounts	**ID: #14**
Stakeholders and goals: Moderator – To search for user accounts	
Description: Moderator inputs keywords to search for user accounts	
Actors: Moderator	
Trigger: When the Moderator uses the search feature	
Pre-condition: Moderator is logged in	
Main flow: 1.Moderator navigates to user management. 2.Moderator enters search keywords. 3.System displays matching user accounts. 4.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 3a. No results i. System displays “No user found.”	

Name: Moderator - View User Accounts	**ID: #15**
Stakeholders and goals: Moderator – To view user account details	
Description: Moderator selects a user account to view details	
Actors: Moderator	
Trigger: When the Moderator clicks on a user profile	
Pre-condition: Moderator is logged in and has searched for users	
Main flow: 1.Moderator clicks on a user from the search results. 2.System displays full user details. 3.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 2a. User data not found i. System displays error message.	

Name: Moderator - Suspend User Account	**ID: #16**
Stakeholders and goals: Moderator – To suspend users	
Description: Moderator suspends a user to restrict access	
Actors: Moderator	
Trigger: When the Moderator clicks the suspend button	
Pre-condition: Moderator is logged in and has selected a user	
Main flow: 1.Moderator clicks “Suspend” on the user profile. 2.System prompts for confirmation. 3.Moderator confirms. 4.System suspends the user and updates status. 5.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 4a. User already suspended i.System displays “User is already suspended.”	

Name: Unsuspend User Account	**ID: #17**
Stakeholders and goals: Moderator – To unsuspend users	
Description: Moderator reactivates a suspended user	
Actors: Moderator	
Trigger: When the Moderator clicks the unsuspend button	
Pre-condition: User account is suspended	
Main flow: 1.Moderator clicks “Unsuspend.” 2.System prompts for confirmation. 3.Moderator confirms. 4.System updates user status to active. 5.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 4a. User not suspended i.System displays “User is not suspended.”	

Name: Create Leaderboard	**ID: #18**
Stakeholders and goals: Moderator – To create leaderboard	
Description: Moderator creates a new leaderboard	
Actors: Moderator	
Trigger: When the Moderator navigates to leaderboard creation	
Pre-condition: Moderator is logged in	
Main flow: 1.Moderator navigates to leaderboard section. 2.Moderator clicks “Create Leaderboard.” 3.Moderator enters details. 4.System saves and displays the leaderboard. 5.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 4a. Missing details i.System prompts to fill required fields.	

Name: Update Leaderboard	**ID: #19**
Stakeholders and goals: Moderator – To update leaderboard data	
Description: Moderator updates existing leaderboard entries	
Actors: Moderator	
Trigger: When the Moderator selects an existing leaderboard	
Pre-condition: Leaderboard exists	
Main flow: 1.Moderator selects leaderboard. 2.Edits rank or data. 3.Clicks “Save.” 4.System updates leaderboard. 5.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 3a. Invalid input i.System highlights errors.	

Name: View Leaderboard	**ID: #20**

Stakeholders and goals: Moderator – To view leaderboard status
Description: Moderator views current leaderboard
Actors: Moderator
Trigger: When the Moderator accesses the leaderboard page
Pre-condition: Leaderboard exists
Main flow: 1.Moderator opens leaderboard page. 2.System displays current rankings. 3.End.
Sub- flow: None.
Alternative/Exceptional flow: 2a. No leaderboard i.System displays “No leaderboard available.”

Name: Create Event	**ID: #21**
Stakeholders and goals: Moderator – To create events for users	
Description: Moderator creates an event for users to join	
Actors: Moderator	
Trigger: When the Moderator navigates to the event section	
Pre-condition: Moderator is logged in	
Main flow: 1.Moderator clicks “Create Event.” 2.Inputs event details. 3.Clicks “Submit.” 4.System saves and displays the event. 5.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 3a. Missing required fields i. System prompts completion.	

Name: Moderator - Update Event	**ID: #22**

Stakeholders and goals: Moderator – To update existing events
Description: Moderator edits an event's information
Actors: Moderator
Trigger: When the Moderator selects an event
Pre-condition: Event exists
Main flow: 1.Moderator selects event. 2.Edits information. 3.Clicks "Save." 4.System updates the event. 5.End.
Sub- flow: None.
Alternative/Exceptional flow: 3a. Invalid data i.System displays error.

Name: Moderator - View Event	**ID: #23**
Stakeholders and goals: Moderator – To view current event status	
Description: Moderator checks details of existing events	

Actors: Moderator
Trigger: When the Moderator accesses the event page
Pre-condition: Event exists
Main flow: 1.Moderator opens event section. 2.Clicks on an event. 3.System displays event details. 4.End.
Sub- flow: None.
Alternative/Exceptional flow: 3a. Event not found i.System displays “Event not available.”

Name: Moderator - Delete Event	**ID: #24**
Stakeholders and goals: Moderator – To delete outdated events	
Description: Moderator removes an existing event	
Actors: Moderator	
Trigger: When the Moderator clicks delete on an event	
Pre-condition: Event exists	
Main flow: 1.Moderator clicks “Delete” on event. 2.System prompts for confirmation. 3.Moderator confirms. 4.System removes the event. 5.End.	
Sub- flow: None.	
Alternative/Exceptional flow: 3a. Event deletion fails i.System displays an error message.	

Name: Moderator - Reset Password	**ID: #25**

Stakeholders and goals: Moderator – To reset password
Description: Moderator resets their password for security
Actors: Moderator
Trigger: When the Moderator clicks “Forgot Password” or accesses settings
Pre-condition: Moderator account exists
Main flow: 1.Moderator clicks “Reset Password.” 2.Enters email or security code. 3.System sends a reset link. 4.Moderator follows link and sets new password. 5.System confirms update. 6.End.
Sub- flow: None.
Alternative/Exceptional flow: 3a. Invalid email/code i. System displays error. 4a. Weak password i.System prompts stronger password.

4.3.4 Business User Use Case Description

Name: Business User - Log In	**ID: #26**
Stakeholders and goals: Business User - To login to the system	
Description: Business User inputs their credentials and clicks on the “Log In” button to log in to the system.	
Actors: Business User	
Trigger: When the Business User enters login details and clicks on the log in button.	
Pre-condition: The Business User has been successfully created	
Main flow: 1. Business User launches the app. 2. System displays for username, and password. 3. Input username, password. 4. Business User clicks the “Log In” button. 5. The system validates the username, password, and user profile. 6. Business User is redirected to the main page. 7. End.	
Sub- flow: None.	
Alternative/Exceptional flow: 5.1a. Blank input: System prompts “Please fill up username/password.” 5.1b. Repeat steps 3 and 4. 5.2a. Invalid login: System display message “Invalid username/password. Please try again.” 5.2b. Business User repeats steps 3 and 4.	

Name: Business User - Logout	**ID: #27**
Stakeholders and goals: Business User - To logout from the system	
Description: Business User clicks on the “Logout” button and logs out from the system.	
Actors: Business User	
Trigger: When the Business User clicks on the logout button.	
Pre-condition: Business User is logged in.	
Main flow: 1. Business User clicks on the “Logout” button. 2. System logs out the Business User. 3. Redirected to the login page. 4. End.	
Sub- flow: None.	
Alternative/Exceptional flow: None.	

Name: Business User - Create Service	**ID: #28**
Stakeholders and goals: Business User - To create services so users can engage with me.	
Description: Business User clicks on “Create Services” button, inserts a new service details. System validates the input and registers into the system.	
Actors: Business User	
Trigger: When Business User clicks the create new service button.	
Pre-condition: 1. Business User is logged in. 2. Business User is currently on their service listing page.	
Main flow: 1. Business User clicks on “Create Services” button. 2. System displays a dialog for creating new services.	

3. Business User inputs the service details.
4. Business User clicks “Create” button.
5. System displays “Service has successfully been created."
6. Business User clicks the “Ok” button.
7. System closes the dialog.
8. End.

Sub-flow: None

Alternative/Exceptional flows:
5.1a. Blank input: System prompts “Please fill up all fields.”
5.1b. Business User repeats steps 3 and 4.

5.2a. Database Error: System prompts “Failed to create a new service. It looks like a server error. Please try a while later!”
5.2b. Business User repeats steps 1 to 4.

Name: Business User - Update Service	**ID: #29**

Stakeholders and goals: Business User - To update the details of my service.

Description: Business User clicks on “Update Details” button, and the system displays the edit details page. Business User inputs the edited details, clicks “update” button, and the system updates the details accordingly.

Actors: Business User

Trigger: When Business User clicks “Update Details” button.

Pre-condition:
1. Business User is logged in.
2. Business User is currently viewing the particular service.

Main flow:

1. Business User clicks on “Update Details” button on the service.
2. System displays the editing field with previous details.
3. Business User inputs the updated details
4. Business User clicks “updated” button.
5. System updates the service details and displays “Service details are successfully updated.”

6. Business User clicks “Done” button.
7. System redirects Business User to the service listing page.
8. End.

Sub-flow: None

Alternative/Exceptional flows:
5.1a. Blank input: System prompts “Please fill in all fields.”
5.1b. Business User repeats steps 3 and 4.

5.2a. Database Error: System prompts “Failed to update the service. It looks like a server error. Please try a while later!”
5.2b. Business User repeats steps 1 to 4.

Name: Business User - Delete Service	**ID: #30**
Stakeholders and goals: Business User - To remove the service	
Description: Business User clicks “Delete” button on the service. The system verifies the action, and removes it from the database.	
Actors: Business User	
Trigger: When Business User clicks “Delete” button.	
Pre-condition: 1. Business User is logged in. 2. Business User is currently viewing the particular service.	
Main flow: 1. Business User clicks “Delete” button on the service. 2. System displays “Are you sure? You are about to delete this service.” 3. Business User clicks on “Yes” button for confirmation. 4. System deletes the service from the database. 5. System prompts “Service successfully deleted”, and redirects the Business User to the service listing page. 6. End.	
Sub-flow: None	

Alternative/Exceptional flows: 3.1a. Business User clicks on “No” button. 3.1b. System redirect the Business User to the service listing page. 4.1a. Database Error: System prompts “Error occurred while deleting service from the database. Please try again.” 4.1b. Business User repeats steps 1 to 3.

Name: Business User - View User Performance Statistics	**ID: #31**
Stakeholders and goals: Business User - To view the details of user performance statistics	
Description: Business User clicks on “View” button of the user, and the system displays its details.	
Actors: Business User	
Trigger: When Business User clicks the “View” button.	
Pre-condition: 1. Business User is logged in. 2. Business User is currently on the user account.	
Main flow: 1. Business User clicks on the “View” button assigned to the user account. 2. System displays selected user data. 3. End.	
Sub-flow: None.	
Alternative/Exceptional flows: None.	

Name: Business User - Create Post	**ID: #32**
Stakeholders and goals: Business User - To create posts so users can learn more tips or gain fitness advices	
Description: Business User clicks on “Create Post” button, inserts the new post details. System validates the input and registers into the system.	
Actors: Business User	
Trigger: When Business User clicks the create post button.	
Pre-condition: 1. Business User is logged in. 2. Business User is currently on their post page.	
Main flow: 1. Business User clicks on “Create Post” button. 2. System displays a dialog for creating new post. 3. Business User inputs the post details. 4. Business User clicks “Create” button. 5. System displays “Post has successfully been posted." 6. Business User clicks the “Ok” button. 7. System closes the dialog. 8. End.	
Sub-flow: None	
Alternative/Exceptional flows: 5.1a. Blank input: System prompts “Please fill up all fields.” 5.1b. Business User repeats steps 3 and 4. 5.2a. Database Error: System prompts “Failed to create a new post. It looks like a server error. Please try a while later!” 5.2b. Business User repeats steps 1 to 4.	

Name: Business User - Delete Post	**ID: #33**
Stakeholders and goals: Business User - To remove the post	
Description: Business User clicks “Delete” button on the post. The system verifies the action, and removes it from the database.	
Actors: Business User	
Trigger: When Business User clicks “Delete” button.	
Pre-condition: 1. Business User is logged in. 2. Business User is currently viewing the particular post.	
Main flow: 1. Business User clicks “Delete” button on the service. 2. System displays “Are you sure? You are about to delete this post.” 3. Business User clicks on “Yes” button for confirmation. 4. System deletes the service from the database. 5. System prompts “Post successfully deleted”, and redirects the Business User to the post page. 6. End.	
Sub-flow: None	
Alternative/Exceptional flows: 3.1a. Business User clicks on “No” button. 3.1b. System redirect the Business User to the post page. 4.1a. Database Error: System prompts “Error occurred while deleting post from the database. Please try again.” 4.1b. Business User repeats steps 1 to 3.	

Name: Business User - Reset Password	**ID: #34**
Stakeholders and goals: Business User - To reset their password in case they forgot their own password.	
Description: Business User clicks on the “Forgot Password” button on the log in page. System verifies the email and send a password reset link to their email	
Actors: Business User	
Trigger: When Business User clicks “Delete” button.	
Pre-condition: 1. Business User is on the login page. 2. Business User has a registered account with the email in the system.	
Main flow: 1. Business User clicks the “Forgot Password” button. 2. The system displays the reset password page asking for an email. 3. Business Users input their registered email address. 4. Click on “Submit” button 5. The system verifies the email and sends a password reset email. 6. Business Users click on the reset link and create a new password. 7. Business User input and confirm the new password. 8. Click on the “Reset” button. 9. System update the password and display “Password successfully changed” 10. Business User can now log in using the new password 11. End.	
Sub-flow: None	
Alternative/Exceptional flows: 5.1a. Database Error: System prompts “Email not registered. Please check and try again!” 5.1b. Re-enter the email and repeat steps 3 and 4. 7.1a. System prompts “Passwords do not match, please try again.” 7.1b. Business User repeats steps 9.	

4.3.5 User Use Case Description

Name: User **-** Login	**#ID: 35**
Stakeholder and goals: User – Login to the system.	
Description: A user wants to access their account.	
Actors: User and Premium User	
Trigger: When the user enters login details and clicks on the login button.	
Prerequisite/ Preconditions: - Users must not be logged in. - User account must be registered.	
Normal Flow: 1. Enter email and password 2. Click on “Login” 3. System redirects to the homepage dashboard.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1. Invalid credentials: An error message will prompt to request user to re-enter a valid credential 2. User account is locked - reset password.	

Name: User - Logout	**#ID: 36**
Stakeholder and goals: User – Logout from the system.	
Description: A user wants to secure their session.	
Actors: User and Premium User	
Trigger: When the user click on the “Logout” Button	
Prerequisite/ Preconditions: - Users must be logged in.	
Normal Flow: 1. The user clicks “Logout”. 2. System ends the session and redirects to the login page	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1. Session timeout error	

Name: User - Reset Password	**#ID: 37**
Stakeholder and goals: User – Reset their password.	
Description: A user wants to reset a forgotten or expired password.	
Actors: User and Premium User	
Trigger: When the user account is locked out.	
Prerequisite/ Preconditions: - User knows email or username	
Normal Flow: 1. Click “Forgot Password” 2. Enter email 3. The system sends a reset link. 4. The user sets a new password.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1. Email not registered 2. Link expired	

Name: User - View Fitness Summary	**#ID: 38**
Stakeholder and goals: User – View workout and fitness progress over time.	
Description: The user wants to monitor their fitness performance by accessing a visual summary of workout data (e.g., calories burned, steps taken, workout frequency).	
Actors: User and Premium User	
Trigger: User navigates to the fitness summary section in the app.	

Prerequisite/ Preconditions:
- Users must be logged in to the WiseWorkout app.
  User must have existing training data recorded in the system.

Normal Flow:
1. The user logs into the WiseWorkout app.
2. The user navigates to the “Summary” section from the main menu.
3. The app displays fitness data in charts or tables, filterable by day, week, or month.
4. User views and interprets trends and progress in fitness activity.

Sub-Flow:
Nil

Alternate/ Exception Flow:
1. No Data Available – App displays a message like “No fitness data available. Start a workout to see your progress.”

Name: User - Upload Fitness Data	**#ID: 39**

Stakeholder and goals:
User – Sync fitness data from external devices (e.g., smartwatches, fitness bands, mobile trackers) to keep workout records up to date.

Description:
Users want to upload and sync fitness data from over 90 supported wearable and mobile sources using the WiseWorkout app.

Actors: User and Premium User

Trigger: Users want to sync and update training information.

Prerequisite/ Preconditions:
- Bluetooth or Wi-Fi Enabled
- Devices must have Bluetooth (preferably BLE) or Wi-Fi turned on.
- Permissions Granted
- Apps must have necessary permissions: Bluetooth, location, internet access.
- Device Discoverability
- Wearable must be visible/discoverable for pairing.
- Protocol Compatibility
- Both devices should support a compatible communication protocol (BLE, Wi-Fi Direct, WebSockets).

- App Installed and Running
- WiseWorkout must be installed and actively running on both devices or in background mode.
-
- Pairing and Authorization Completed
- Devices must be paired and authenticated for secure sync.
- Data Format Agreement
- Devices and apps must agree on supported data structures (e.g., JSON format for steps, heart rate, etc.).
- App Installed and Running
- Required syncing apps must be installed and active on both devices (or running as a background service).
- Pairing and Authorization Completed
- Devices must be paired and authenticated if secure data exchange is needed.
- Data Format Agreement
- Both ends must agree on a common data format and sync structure (e.g., JSON fields for steps, heart rate).

Normal Flow:

1. Users launch the WiseWorkout app on their mobile device.
2. The app verifies that Bluetooth or Wi-Fi is enabled.
3. App scans for compatible wearable devices.
4. The user selects a device to sync with.
5. Devices connect and authenticate.
6. Fitness data is pulled from the external device.
7. Data is stored locally and optionally synced to the cloud.
8. UI updates to reflect new training stats.

Sub-Flow:

Nil

Alternate/ Exception Flow:

- Sync Failed – Connection timeout, transmission error, or API error.
- Device Not Connected – Wearable not paired, powered off, or out of range.
- Permission Denied – Required system permissions not granted by user.
- Unsupported Format – Device returns data in an unrecognized structure.

Name: User **-** View Workout History	**#ID: 40**
Stakeholder and goals: User – View a list of completed past workouts to track performance and consistency.	
Description: The user wants to review details of previous exercise sessions, including type, duration, and stats.	
Actors: User and Premium User	
Trigger: User selects the "Workout History" option from the app menu.	
Prerequisite/ Preconditions: - Users must be logged in. - Users must have existing workout data stored in the app.	
Normal Flow: 1. The user logs in to the WiseWorkout app. 2. The user navigates to the "Workout History" section. 3. The app displays a list of past workouts with relevant details (e.g., date, type, duration). 4. The user browses and optionally selects an entry to view more information.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1. No History Found – App displays a message: “You haven’t logged any workouts yet. Start your first workout to begin tracking.”	

Name: User **-** View workout progress	**#ID: 41**
Stakeholder and goals: User - Monitor progress toward fitness goals through visual insights.	
Description: The user checks how well they are progressing toward their fitness or workout goals by reviewing visual data.	
Actors: User and Premium User	
Trigger: When the user click on the “view workout progress” button.	
Prerequisite/ Preconditions: - Users must have an active workout program: The user needs to have goals or metrics set, which are tracked by the app (e.g., running a certain number of kilometers or lifting a specific amount of weight). - The user must have completed at least one workout: There must be data points to compare progress against, so the user should have logged past workouts.	
Normal Flow: 1. The user navigates to the Progress tab in the app. 2. The user clicks the Progress tab to view progress data. 3. Upon clicking, the app displays the user’s progress data. 4. The app displays the following: 4.1. Overview metric (general progress toward fitness goals). 4.2. Calories burned (graph or bar showing calories burned over time). 4.3. Sub-view metric (detailed metrics such as distance run or total exercise time).	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1. Insufficient data to plot for weekly or monthly progress - system will display a placeholder visualization with message to motivate users to clock more sessions.	

Name: User **-** Share to Social Media	**#ID: 42**
Stakeholder and goals: User – Share training results to IG/FB/TikTok.	
Description: User posts results to external social media.	
Actors: User and Premium User	
Trigger: - The user completes the workout. - User wants to share workout results on social media.	
Prerequisite/ Preconditions: - Users must have an active account on the app. - Users must have linked social media accounts (IG/FB/TikTok). - User must be logged in.	
Normal Flow: 1. Tap "Share to Social Media" button → *User action in workout summary screen → To begin sharing process*. 2. Select platform → *User chooses from IG, FB, TikTok → To target desired audience*. 3. Choose training result → *User picks a completed session from the list → To share a specific performance*. 4. Enter caption (optional) → *User types a message → To provide personal context*. 5. Tap "Post" button → *User initiates upload → To share on chosen platform*. 6. System confirms success → *Confirmation message appears → To notify that post was shared*.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 3a. Auth failure with social media platform → System detects token/authentication issue → Display error message – “Unable to connect to IG/FB/TikTok.” → Prompt to re-authenticate or retry – User can log in again or cancel.	

Name: User **-** Search User	**#ID: 43**
Stakeholder and goals: User – Connect with others for community engagement or training support.	
Description: User searches for other users within the app using name or interest keywords to initiate connections.	
Actors: User and Premium User	
Trigger: User click on the “Search” button.	
Prerequisite/ Preconditions: - Users must be logged in. - Search feature must be enabled and accessible.	
Normal Flow: 1. Click on the "Search" icon – to access the user search page. 2. Type in keywords – such as name, username, or interests (e.g., "yoga", "HIIT"). 3. System displays search results – filtered based on matching criteria. 4. The user scrolls through the list – to browse potential matches. 5. Click on a user profile – to view more details and send connection/follow requests.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 3a. No user found: - Display message: "No matching users found." - Suggest possible corrections (e.g., "Try a different name or interest.")	

Name: User - Add Friend	**#ID: 44**
Stakeholder and goals: User – Allow them to follow other user	
Description: User send a friend request	
Actors: User and Premium User	
Trigger: User click on the “Add Friend” button.	
Prerequisite/ Preconditions: - Users must be logged in. - Target User must exist.	
Normal Flow: 1. Search for the user 2. Click on the user profile 3. Click on Add Friend	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1a. No user found: - Display message: "No matching users found." - Suggest possible corrections (e.g., "Try a different name or interest.")	

Name: User - Remove Friend	**#ID: 45**
Stakeholder and goals: User – Allow them to remove other user	
Description: User remove someone from their friends list	
Actors: User and Premium User	
Trigger: User wants to connect with other individuals on the platform.	
Prerequisite/ Preconditions: - Users must be logged in. - Already Friends	
Normal Flow:	

1. Search for the user 2. Click on the user profile 3. Click on Remove Friend
Sub-Flow: Nil
Alternate/ Exception Flow: 3a. Database Error: System prompts “User can be removed. Please try again later!”

Name: User - Accept Friend Request	**#ID: 46**
Stakeholder and goals: User – Allow them to accept who they want to be friends with.	
Description: User accept pending friend request	
Actors: User and Premium User	
Trigger: User clicks on the “Accept” button.	
Prerequisite/ Preconditions: - Users must be logged in. - Have a pending request.	
Normal Flow: 1. See a friend request notification 2. Click “Accept”	
Sub-Flow: Nil	
Alternate/ Exception Flow: Nil	

Name: User - Decline Friend Request	**#ID: 47**
Stakeholder and goals: User – Allow them to reject who they do not want to be friends with.	

Description: User decline pending friend request
Actors: User and Premium User
Trigger: User clicks on the “Reject” button.
Prerequisite/ Preconditions: - Users must be logged in. - Have a pending request
Normal Flow: 1. See a friend request notification. 2. Click “Reject”.
Sub-Flow: Nil
Alternate/ Exception Flow: Nil

Name: User - View Friend List	**#ID: 48**
Stakeholder and goals: User – Allow them to manage friend request	
Description: Allow them to view the friend list.	
Actors: User and Premium User	
Trigger: When the user are on their friend list page	
Prerequisite/ Preconditions: - Users must be logged in. - User must have friends	
Normal Flow: 1. On the friend list page	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1a. Database Error: System prompts “Error, please try again later!”	

Name: User - View Friend's Feed to like	**#ID: 49**
Stakeholder and goals: User – Allow them to interact with other user	
Description: Users can choose to like a friend's post.	
Actors: User and Premium User	
Trigger: When the user are on their friend's post	
Prerequisite/ Preconditions: - Users must be logged in. - User must have friends - User is viewing the post	
Normal Flow: 1. User see the post. 2. User click on the like button.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1a. Database Error: System prompts "Error, please try again later!"	

Name: User - View Friend's Feed to comment	**#ID: 50**
Stakeholder and goals: User – Allow them to interact with other user	
Description: Users can choose to comment on a friend's post.	
Actors: User and Premium User	
Trigger: When the user are on their friend's post	
Prerequisite/ Preconditions: - Users must be logged in. - User must have friends - User is viewing the post	
Normal Flow:	

1. User see the post. 2. User click on the comment button. 3. User comment on something.
Sub-Flow: Nil
Alternate/ Exception Flow: 3a. Database Error: System prompts “Fail to post comment, please try again later!”

Name: User - View Event Calendar	**#ID: 51**
Stakeholder and goals: User – Allow them see the events available	
Description: Users see available events.	
Actors: User and PremiumUser	
Trigger: When the user clicks on the “View Event” button.	
Prerequisite/ Preconditions: - Users must be logged in. - There must be events scheduled	
Normal Flow: 1. Open Calendar	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1a. Database Error: System prompts “Error, please try again later!” 1b. No events available	

Name: User - Join Event	**#ID: 52**
Stakeholder and goals: User – Allow them to join a event	
Description: Users can choose to join the events	
Actors: User and PremiumUser	

Trigger: When the user clicks on the “Join” button.
Prerequisite/ Preconditions: - Users must be logged in. - There must be an event scheduled.
Normal Flow: 1. User is on the event page 2. User click on the event they want to join 3. Click on “Join” button
Sub-Flow: Nil
Alternate/ Exception Flow: 3a. Database Error: System prompts “Error, please try again later!” 3b. Event is full or closed.

Name: User - Shortlist Event	**#ID: 53**
Stakeholder and goals: User – Allow them to shortlist event	
Description: Users can choose to save the event they want	
Actors: User and PremiumUser	
Trigger: When the user clicks on the “Shortlist” button.	
Prerequisite/ Preconditions: - Users must be logged in. - There must be an event scheduled.	
Normal Flow: 1. User is on the event page 2. User click on the event they want to shortlist 3. Click on “Shortlist” button	
Sub-Flow: Nil	
Alternate/ Exception Flow: 3a. Database Error: System prompts “Error saving, please try again later!”	

3b. Event is full or closed.

Name: User - Create a Team (Up to 4 pax)	**#ID: 54**
Stakeholder and goals: User – Allow them to create a team	
Description: Users can choose to create their own team.	
Actors: User and PremiumUser	
Trigger: When the user clicks on the “Create” button.	
Prerequisite/ Preconditions: - Users must be logged in. - Must not be in any teams yet	
Normal Flow: 1. User click on the “Create” Button 2. A team is created	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1a. Database Error: System prompts “Error creating team, please try again later!” 1b. Already in an existent team.	

Name: User **-** Send Team Invite (1 max)	**#ID: 55**
Stakeholder and goals: User – Invite friends to team**.**	
Description: User sends a single invite to a friend to join a group workout.	
Actors: User and Premium User	
Trigger: When the user chooses to invite a friend.	
Prerequisite/ Preconditions: User must be team owner and has not used their one-time invite.	
Normal Flow: 1.User opens the group workout section. 2.System displays the invite option. 3.User selects “Invite a Friend.” 4.System checks if the user is eligible (invite limit not reached). 5.User enters friend’s contact. 6.System sends invites and confirms success. 7.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 5a. If the limit is reached: System displays “Invite limit reached. Upgrade to Premium for unlimited invites.” 5b. End.	

Name: User **-** Join Team	**#ID: 56**
Stakeholder and goals: User – Join an existing team.	
Description:. User accepts invitation.	
Actors: User and Premium User	
Trigger: When a user selects to join a team**.**	
Prerequisite/ Preconditions: User is logged in and not currently in a team.	
Normal Flow: 1.User navigates to team section. 2.Views list of available teams. 3.Selects a team to join. 4.Confirms selection. 5.System adds user to team and displays success. 6.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 4a. If already in a team: Show message “Leave your current team to join a new one.”	

Name: User **-** Leave Team	**#ID: 57**
Stakeholder and goals: User – Exit current team.	
Description: User leaves their current team.	
Actors: User and Premium User	
Trigger: When user clicks “Leave Team” option.	
Prerequisite/ Preconditions: User is logged in and part of a team.	
Normal Flow: 1.User goes to team page. 2.Clicks “Leave Team”. 3.System asks for confirmation. 4.User confirms. 5.System removes user from team and confirms. 6.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 5a. If user clicks “Leave Team” button, but a network error occurs: System displays message: “Unable to process request. Please check your connection and try again.”	

Name: User **-** Browse Training Programs	**#ID: 58**
Stakeholder and goals: User – Explore available programs.	
Description: User views a list of training programs.	
Actors: User and Premium User	
Trigger: When user opens the training program section.	
Prerequisite/ Preconditions: User must be logged in.	
Normal Flow: 1.User navigates to “Training Programs.” 2.System displays available programs with filters. 3.User browses through list. 4.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 2a. If no training programs are available, the system displays: “No programs found. Please check back later.”	

Name: User **-** Rate and Review Programs	**#ID: 59**
Stakeholder and goals: User – Provide feedback.	
Description: User rates and writes a review.	
Actors: User and Premium User	
Trigger: When user finishes or views a training program.	
Prerequisite/ Preconditions: User is logged in.	
Normal Flow: 1.User opens a completed program. 2.Clicks “Rate & Review”. 3.Inputs rating and optional comment. 4.Submits feedback. 5.System saves review. 6.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 4a.If the user submits the form without a rating, the system shows: “Please select a rating before submitting.” 4b.If a network error occurs during submission, the system shows: “Submission failed. Please try again.”	

Name: User **-** Save Training Program to Favorites (Limit 3)	**#ID: 60**
Stakeholder and goals: User – Bookmark programs.	
Description: Save training programs to favorites list.	
Actors: User	
Trigger: When user clicks "Save to Favorites"**.**	
Prerequisite/ Preconditions: User is logged in and has saved fewer than 3 programs.	
Normal Flow: 1.User clicks "Save". 2.System checks favorite limit. 3.Saves to list and confirms. 4.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 2a. If at limit: System shows “Save limit reached. Upgrade to Premium for unlimited saves.”	

Name: User **-** View Profile	**#ID: 61**
Stakeholder and goals: User – View personal info.	
Description: User views their profile.	
Actors: User and Premium User	
Trigger: When user opens profile.	
Prerequisite/ Preconditions: User is logged in.	
Normal Flow: 1.User navigates to profile. 2.System displays profile details. 3.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1a. If the system fails to load profile (e.g., network error): System shows error message “Unable to load profile. Please try again later.”	

Name: User **-** Update Profile	**#ID: 62**
Stakeholder and goals: User – Edit user data.	
Description:User updates basic details.	
Actors: User and Premium User	
Trigger: When user clicks "Edit Profile" button.	
Prerequisite/ Preconditions: User is logged in.	
Normal Flow: 1.User clicks “Edit”. 2.Changes details. 3.Clicks "Save". 4.System updates info and confirms. 5.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 3a. If user leaves required fields blank or inputs invalid data: System shows validation error and highlights the incorrect fields. 4a. If the system fails to update data (e.g., server issue): Show message “Update failed. Please try again later.”	

Name: User **-** Update Health Info	**#ID: 63**
Stakeholder and goals: User – Edit health metrics.	
Description: User updates weight, height, BMI, etc.	
Actors: User and Premium User	
Trigger: When user selects "Health Info".	
Prerequisite/ Preconditions: User is logged in.	
Normal Flow: 1.User navigates to health settings. 2.Inputs new data. 3.Clicks “Save”. 4.System updates data and confirms. 5.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 2a. If user enters invalid data (e.g., negative weight): System displays error and prompts correction. 4a. If system storage fails: System displays “Unable to save changes. Try again.”	

Name: User **-** View Ads/Offers	**#ID: 64**
Stakeholder and goals: User – See promotions.	
Description: User views sponsor content.	
Actors: User and Premium User	
Trigger: When opening the home or explore page.	
Prerequisite/ Preconditions: User is logged in.	
Normal Flow: 1.User scrolls app. 2.Ads or sponsor content shown. 3.User can click to learn more. 4.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 3a. If clicking the ad fails (e.g., link broken): System shows “Unable to open offer. Please try again later.”	

Name: User **-** Provide Feedback	**#ID: 65**
Stakeholder and goals: User – Suggest improvements.	
Description:User submits app feedback.	
Actors: User and Premium User	
Trigger: User selects the “Provide Feedback” option.	
Prerequisite/ Preconditions: User is logged in.	
Normal Flow: 1.User opens feedback form. 2.Writes message. 3.Clicks “Send”. 4.System confirms submission. 5.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 2a. If feedback message is blank: System prompts “Feedback cannot be empty.” 4a. If system submission fails: Show “Feedback not sent. Please retry later.”	

Name: User **-** View Leaderboard	**#ID: 66**

Stakeholder and goals: User – Check global rank.
Description: User checks the leaderboard.
Actors: User and Premium User
Trigger: When user navigates to leaderboard section.
Prerequisite/ Preconditions: User is logged in.
Normal Flow: 1.User opens leaderboard tab. 2.System displays global ranks. 3.End.
Sub-Flow: Nil
Alternate/ Exception Flow: 2a. If leaderboard fails to load (e.g., network error): System shows “Unable to retrieve leaderboard. Try again later.”

Name: User **-** View Subscription Status	**#ID: 67**
Stakeholder and goals: User – Check Premium access.	
Description: User checks subscription type.	
Actors: User and Premium User	
Trigger: When the user opens the subscription page.	
Prerequisite/ Preconditions: User is logged in.	
Normal Flow: 1.User goes to subscription settings. 2.System shows current status (Freemium or Premium). 3.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 2a. If system fails to fetch subscription details: Display message “Unable to retrieve subscription status.”	

Name: User **-** Subscribe to Premium	**#ID: 68**
Stakeholder and goals: User – Upgrade account.	
Description: User subscribes for Premium.	
Actors: User	
Trigger: When user clicks “Upgrade to Premium” button.	
Prerequisite/ Preconditions: User is logged in.	
Normal Flow: 1.User clicks upgrade. 2.Chooses plan and payment method. 3.Submits payment. 4.System verifies and activates Premium. 5.Confirmation message shown. 6.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 3a. If payment info is invalid or incomplete: System shows error message and asks user to correct it. 3b. If payment fails: System notifies user “Payment unsuccessful. Please try again.” 4a. If subscription status fails to update: Display message “Subscription failed. Please contact support.”	

4.3.6 Premium User Use Case Description

Name: Premium User **-** Cancel Premium	**#ID: 69**
Stakeholder and goals: User – End Premium plan.	
Description: User stops future billing.	
Actors: Premium User	
Trigger: When user clicks “Cancel Subscription” button.	
Prerequisite/ Preconditions: User is logged in and must be a Premium subscriber.	
Normal Flow: 1.User goes to subscription page. 2.Clicks “Cancel”. 3.System asks for confirmation. 4.User confirms. 5.System schedules cancellation at period end. 6.End.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 5a. If the cancellation process fails (e.g., server error): System shows message “Cancellation failed. Please try again later.”	

Name: Premium User - Analyze Advanced Metrics	**#ID: 70**
Stakeholder and goals: Premium User – View and analyze detailed performance metrics to track long-term progress.	
Description: Premium users access in-depth analytics such as trends, averages, and comparisons over time.	
Actors: Premium User	
Trigger: User navigates to the “Advanced Metrics” section from the dashboard or profile.	
Prerequisite/ Preconditions: - Users must be logged in. - Users must have an active Premium subscription.	
Normal Flow: 1. Go to “Advanced Metrics” – accessible from dashboard or performance tab. 2. Select time period – choose from predefined ranges (e.g., 7 days, 30 days, custom). 3. System fetches and displays metrics – includes graphs, charts, and numerical data. 4. User views trends and comparisons – analyze progress such as VO2 max, pace, reps, calorie burn. 5. Optionally export/share data – download or share via linked apps (e.g., Google Fit, Apple Health).	
Sub-Flow: Nil	
Alternate/ Exception Flow: 3a. If no data available for selected period: - The system displays a message: “No data found for the selected time range.” - Suggest users to complete more workouts to generate data.	

Name: Premium User - Analyze Workout Intervals	**#ID: 71**
Stakeholder and goals: Premium User – Analyze workout intervals using advanced charts and metrics to optimize performance	
Description: Premium user accesses the “Interval Analysis” feature to break down session data, visualize intervals with performance insights, and optimize future workouts.	
Actors: Premium User	
Trigger: User opens the “Interval Analysis” page after completing a workout session.	
Prerequisite/ Preconditions: - User must be a Premium subscriber - User must have run/workout session data - User must have set preferred metric units (e.g., kg, g, miles, inch, m, km)	
Normal Flow: 1. Open “Interval Analysis” from the app menu or dashboard. 2. Select a specific workout session from the session history. 3. System fetches and displays interval breakdown – includes pace, distance, rest time per segment. 4. The system presents visual analysis – includes bar charts, split tables, and metric highlights. 5. User reviews statistics for performance tuning.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1. If no workout data is available for the selected session:- Display message: “No data found for the selected session. Please ensure data is synced.” 2. If the system fails to fetch interval data: - Display message: “Unable to load interval data. Please try again later.” 3. If the selected metric is in miles: - System converts values to kilometers or meters based on user preference. 4. If the selected metric is in kilometers/meters: - System converts values to miles for users with imperial preference.	

Name: Premium User - Analyze Fitness Trends	**#ID: 72**
Stakeholder and goals: Premium User – Track long-term fitness using charts on power, heart rate, pace, and distance.	
Description: Premium users can view 30+ charts for tracking fitness trends over time, including heart rate, pace, distance, and power data.	
Actors: Premium User	
Trigger: User accesses the “Trends Dashboard” from the app menu.	
Prerequisite/ Preconditions: - Must have multiple sessions logged.	
Normal Flow: 1. Access “Trends Dashboard” from the app menu or dashboard. 2. Select a type of trend to view (e.g., HR, pace, distance, power). 3. View trends – charts are displayed with data for the selected category.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1. If there is no data available: - Display a default dummy chart as a placeholder. - Show message: “No data found. Please complete more sessions to view trends.”	

Name: Premium User - Save Unlimited Programs	**#ID: 73**
Stakeholder and goals: Premium User – Bookmark and save unlimited training programs for future use.	
Description: Premium users can save an unlimited number of training programs to their favorites list for easy access later.	
Actors: Premium User	
Trigger: User views a training program and decides to save it.	
Prerequisite/ Preconditions: - Users must have an active Premium subscription.	
Normal Flow: 1. The user navigates to the "Training Programs" section in the app. 2. Users browse or search for a specific training program they want to save. 3. The user selects the desired training program to view its details. 4. The user clicks on the "Save" icon or "Add to Favorites" button associated with the training program. 5. System confirms the program is added to the "Favorites" list with a brief success message ("Added to Favorites"). 6. The user navigates to the "Favorites" or "Saved Programs" section from the main menu. 7. Users view their saved list of training programs. 8. User selects a saved program from the list to revisit it.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1. If saving fails due to connectivity or server issues: - System displays an error message: “Unable to save this training program. Please try again later.” - Users are not able to save the program until the issue is resolved. 2. If the program is already saved: - System displays a message: “This program is already in your Favorites list.”	

Name: Premium User - Accessed Personalized Plan	**#ID: 74**
Stakeholder and goals: Premium User – Use training custom routines.	
Description: Recommendation for a personalized training plan tapper to user.	
Actors: Premium User	
Trigger:	
Prerequisite/ Preconditions: - Must be a premium user account - Must have existing data for recommendation prompt	
Normal Flow: 1) Premium User logs into their account. 2) The user navigates to the "Personalized Plan" section. 3) The system verifies premium subscriptions and checks if sufficient data exists. 4) The system prompts the user to confirm or update preferences/goals (if applicable). 5) The system generates and displays the personalized training plan based on the latest user data. 6) The user views the plan with details such as workout types, duration, and frequency. 7) The user selects an option to start, save, or edit the plan.	
Sub-Flow: Nil	
Alternate/ Exception Flow: A1: User does not have a premium subscription - The system displays a message prompting the user to upgrade. - The user is redirected to the subscription page. A2: Insufficient data for plan generation - The system notifies the user that more data is needed. - Suggests completing a fitness assessment or logging a few workouts. - Optionally, provides a generic starter plan. A3: Recommendation Engine fails to generate a plan (system error) - The system shows an error message and suggests trying again later. - Logs the error for admin review.	

Name: Premium User - Create Teams with no member limits	**#ID: 75**
Stakeholder and goals: Premium User – Unlimited Members in a team	
Description: Creating a team with unlimited members	
Actors: Premium User	
Trigger: When user clicks on the “Create” button	
Prerequisite/ Preconditions: - Must be a premium user account - Must have existing data for recommendation prompt	
Normal Flow: 1. User click on the “Create” Button 2. A team is created.	
Sub-Flow: Nil	
Alternate/ Exception Flow: 1a. User does not have a premium subscription - The system displays a message prompting the user to upgrade. - The user is redirected to the subscription page. 2a. Database Error: System prompts “Error creating team, please try again later!” 2b. Already in an existent team.	

5. Proposed Plan

5.1 Execution Strategy

Based on our team's extensive experience in software development, we will adopt the SCRUM methodology to efficiently manage deliverables while maintaining a responsive and adaptive approach. SCRUM enables iterative development, fosters collaboration, and ensures continuous feedback, allowing us to address challenges promptly and align with project goals. Below, we outline the methods we plan to use, the proposed timeline, and the rationale for selecting SCRUM for this project

5.2 Software Development Methodology

Methods

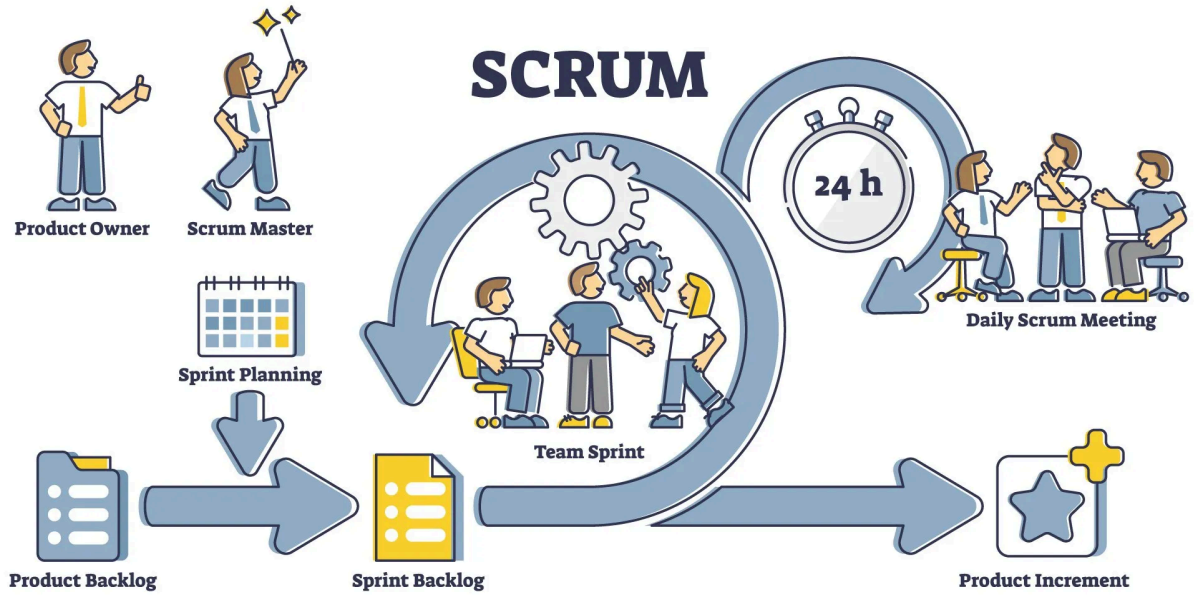


Image taken from ((ProfitableProcesses.com, n.d.)

- Sprint Planning: Conduct weekly sprint planning sessions to define clear, achievable objectives and prioritize tasks based on project requirements.

- Sprint Reviews: At the end of each sprint, conduct reviews to demo deliverables to stakeholders and retrospectives to reflect on process improvements.

- Backlog Grooming: Regularly refine the product backlog to ensure clarity, relevance, and prioritization of tasks.

Proposed Timeline and Duration

- 1 Week per sprint that will be.

- Total Project Duration: Approximately 20 weeks, starting from 5th April 2025 till 19 Aug 2025. We will have about 18 sprints, including the first meeting, and intermediate meeting before any major milestone.

- Milestone

	Date	Milestone/ Activity
Milestone 1	5-Apr-25	Project Briefing; Team Formation; Start of Term 1; Start of Session 1
Milestone 2	7-Apr-25	Submit: Team Members & Leader Details, Website Link, Reflective Diary 1
Milestone 3	2-May-25	Submit: Project Requirement Documentation (includes Literature Review, Tools, Diagrams
Milestone 4	15-May-25	Submit: System Requirement Specification (SRS)
Milestone 5	25-May-25	Submit: Technical Design Manual with Functional Prototype
Milestone 6	10-Jun-25	Submit: System Prototype with Basic Functionalities, Project Progress Report
Milestone 7	21-Jun-25	Milestone: Prototype Demonstration to Supervisor & Assessor
Milestone 8	12-Jul-25	Submit: Individual Functional Modules, Centralized DB, Module Test Plan
Milestone 9	1-Aug-25	Submit: Integrated Module Functionalities & Draft Final Documents
Milestone 10	10-Aug-25	Submit: Final Project Documentation, Project Video, Presentation to Supervisor
Milestone 11	23-Aug-25	Final Presentation & Demonstration to Assessor and Supervisors (Final Milestone)

5.3 Risk Analysis

		Risk Elements		
Risk Category	Identified Risk	Impact	Probability	Risk Rating
Communication	1. Miscommunication between developers and fitness experts	M	L	M
	2. Delay in response from external partners or sponsors	M	M	M
	3. Ambiguity in user feedback translation to features	M	M	M
Resources	1.Limited budget for premium features or sensors	M	L	L
	2. Inadequate number of developers or testers	M	M	M
	3. Hardware limitations in users' phones/wearables	M	M	M
Technical	1.Integration issues with third-party devices (e.g., wearables)	M	M	M

	2. Incompatibility with certain Android/iOS versions	M	M	M
	3. Data syncing delays or failures	M	M	M
	4. Bugs affecting performance across screen sizes	M	H	H
User	1. Low user engagement due to complex UI	H	L	M
	2. Users not inputting accurate health data	M	M	M
	3. Privacy concerns discourage registration	H	M	H
	4. User dissatisfaction from unmet fitness goals	M	M	M
Maintenance	1. Frequent updates required due to device API changes	M	M	M
	2. Technical debt from rushed releases	M	M	M

	3. Outdated content (fitness advice, plans)	M	M	M
Security	1. Data leaks of user health info or personal details	H	L	M
	2. Insecure storage of login or fitness data	H	M	H
	3. Unauthorized access to user data	H	M	H
Scalability	1. Backend fails under high traffic (e.g., competitions)	H	M	M
	2. Limitations in scaling real-time features (e.g., leaderboards)	H	M	M
	3. App performance degrades on low-end devices	M	H	H
Reliability	1. App crashes or freezes during workouts	H	L	M
	2. Inconsistent notification delivery (reminders)	M	M	M
	3. Sync failure between devices (watch ↔ phone)	M	M	M

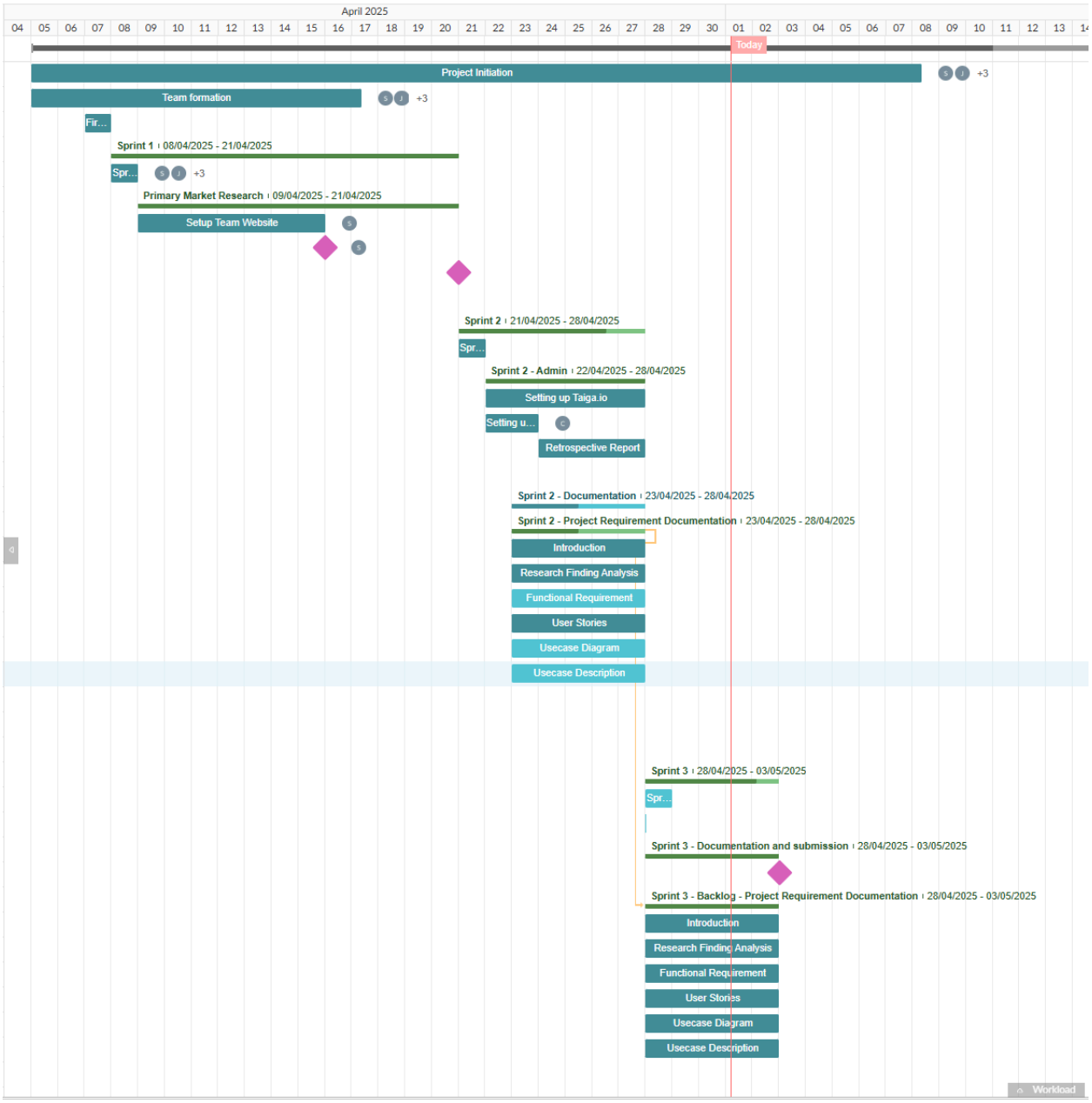
5.4 Proposed Duration and Schedule

FYP-25-S2-09 No status

Gantt chart Board List Calendar Workload People Dashboard

Expand all Collapse all Cascade sorting

	Task name	Assigned	Start date	End date	Duration	Progress	Status
			05/04/202	20/05/202	1080h	80%	
1	Project Initiation	S J +3	05/04/202	08/05/202	800h	100%	Done
2	Team formation	S J +3	05/04/202	17/04/202	296h	100%	Done
3	First meeting kickoff	unassigned	07/04/202	08/04/202	24h	100%	Done
4	Sprint 1		08/04/202	21/04/202	312h	100%	
4.1	Sprint 1 Planning	S J +3	08/04/202	09/04/202	24h	100%	Done
4.2	Primary Market Research		09/04/202	21/04/202	288h	100%	
4.3	Setup Team Website	S Shann...	09/04/202	16/04/202	168h	100%	Done
4.4	Milestone 1 - website complete	S Shann...	16/04/202	16/04/202		100%	Done
4.5	Milestone 2 - Meeting with boss	unassigned	21/04/202	21/04/202		100%	Done
	Add a task | Add a milestone						
5	Sprint 2		21/04/202	28/04/202	168h	79%	
5.1	Sprint 2 Planning	unassigned	21/04/202	22/04/202	24h	100%	Done
5.2	Sprint 2 - Admin		22/04/202	28/04/202	144h	100%	
5.2.1	Setting up Taiga.io	unassigned	22/04/202	28/04/202	144h	100%	Done
5.2.2	Setting up File Sharing System	C CM	22/04/202	24/04/202	48h	100%	Done
5.2.3	Retrospective Report	unassigned	24/04/202	28/04/202	96h	100%	Done
	Add a task | Add a milestone						
5.3	Sprint 2 - Documentation		23/04/202	28/04/202	120h	50%	
5.3.1	Sprint 2 - Project Requirement Documentation		23/04/202	28/04/202	120h	50%	
5.3.1.1	Introduction	unassigned	23/04/202	28/04/202	120h	100%	Done
5.3.1.2	Research Finding Analysis	unassigned	23/04/202	28/04/202	120h	100%	Done
5.3.1.3	Functional Requirement	unassigned	23/04/202	28/04/202	120h	0%	Open
5.3.1.4	User Stories	unassigned	23/04/202	28/04/202	120h	100%	Done
5.3.1.5	Usecase Diagram	unassigned	23/04/202	28/04/202	120h	0%	Open
5.3.1.6	Usecase Description	unassigned	23/04/202	28/04/202	120h	0%	Open
	Add a task | Add a milestone						
	Add a task | Add a milestone						
	Add a task | Add a milestone						
6	Sprint 3		28/04/202	03/05/202	120h	83%	
6.1	Sprint 3 - Planning	unassigned	28/04/202	29/04/202	24h	0%	Open
6.2	Sprint 3 - Admin	unassigned	28/04/202	28/04/202	1m	0%	Open
6.3	Sprint 3 - Documentation and submission		28/04/202	03/05/202	120h	100%	
6.3.1	Milestone 3 - PRD Submission	unassigned	03/05/202	03/05/202		0%	Open
6.3.2	Sprint 3 - Backlog - Project Requirement Documentation		28/04/202	03/05/202	120h	100%	
6.3.2.1	Introduction	unassigned	28/04/202	03/05/202	120h	100%	Done
6.3.2.2	Research Finding Analysis	unassigned	28/04/202	03/05/202	120h	100%	Open
6.3.2.3	Functional Requirement	unassigned	28/04/202	03/05/202	120h	100%	Open
6.3.2.4	User Stories	unassigned	28/04/202	03/05/202	120h	100%	Done
6.3.2.5	Usecase Diagram	unassigned	28/04/202	03/05/202	120h	100%	Open
6.3.2.6	Usecase Description	unassigned	28/04/202	03/05/202	120h	100%	Open
	Add a task | Add a milestone						


April 2025
04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 01 02 03 04 05 06 07 08 09 10 11 12 13 14
Today
Project Initiation
Team formation
Fir...
Sprint 1 | 08/04/2025 - 21/04/2025
Spr...
Primary Market Research | 09/04/2025 - 21/04/2025
Setup Team Website
Sprint 2 | 21/04/2025 - 28/04/2025
Spr...
Sprint 2 - Admin | 22/04/2025 - 28/04/2025
Setting up Taiga.io
Setting u...
Retrospective Report
Sprint 2 - Documentation | 23/04/2025 - 28/04/2025
Sprint 2 - Project Requirement Documentation | 23/04/2025 - 28/04/2025
Introduction
Research Finding Analysis
Functional Requirement
User Stories
Usecase Diagram
Usecase Description
Sprint 3 | 28/04/2025 - 03/05/2025
Spr...
Sprint 3 - Documentation and submission | 28/04/2025 - 03/05/2025
Sprint 3 - Backlog - Project Requirement Documentation | 28/04/2025 - 03/05/2025
Introduction
Research Finding Analysis
Functional Requirement
User Stories
Usecase Diagram
Usecase Description
Workload

5.5 Roles and Responsibility

Name	Roles	Responsibilities	Contact Information
Kang YiYan	Lead Developer, Software Tester, Documentation	Design and code and come out with test cases along with documentation.	ykang009@mymail.sim.edu.sg 9673 8687
Lock Xue Min Shannon	Project Leader, Developer, Documentation	Monitor and delegate the task to members. Design and code the system along with documentation.	xmslock001@mymail.sim.edu.sg 92727237
Yang Jindu	Lead Tester, Front-End Developer, Documentation	Design and code the system. Prepare and run test cases along with documentation	jyang050@mymail.sim.edu.sg 8509 8950
Ching Jun Jie Jay	Lead Developer, Front and back end Developer, Security Engineer	In charge of the overall quality of the project. Design and code the front and back of the system. Run test on system. Implement security function.	Jjjching001@mymail.sim.edu.sg 9137 4722
Teo Choun Meng	Lead Documentation, Software tester, Developer	Manage the documentation of the project. Design and test the system ensuring it meets the requirement.	cmzjteo001@mymail.sim.edu.sg 8668 1440

6. Development Tools

Front-end	
Development Tools	Visual Studio Code/Android Studio, Notepad++
Language	Dart
Framework	Flutter
Description	
Flutter is an open-source framework developed and supported by Google. Frontend and full-stack developers use Flutter to build an application's user interface (UI) for multiple platforms with a single codebase. Flutter simplifies the process of creating consistent, appealing UIs and uses Dart, which is an open-source programming language, also created by Google.	
Back-end	
Development Tools	Visual Studio Code/Android Studio, Notepad++
Language	Javascript/Swift/Kotlin/Java
Description	

Google Firebase is a set of cloud-based development tools that helps mobile app developers build, deploy, and scale their apps. The Realtime Database is Firebase's original database offering. It is a NoSQL database that stores data as JSON objects, allowing for real-time synchronization across connected devices.

Database	
Development tools	MySQL workbench
Language	MySQL
Description	
MySQL is a Relational Database Management System (RDBMS) based on Structured Query Language (SQL). SQL is a language use to create, modify, and extract data from the relational database, as well as control user access to the database. Why MySQL? ● Provide scalability and flexibility ● High performance ● High availability ● Strong data protection ● Management ease	

7. Expected Results

Goal	Expected Result
Personalized fitness	Users get workout plans tailored to their goals, fitness levels, and preferences.
Improved health and fitness	Regular use leads to measurable improvements in strength, endurance, flexibility, or weight loss.
Progress tracking	Users can see trends, history, and milestones (e.g., more reps, longer workouts, weight change).
Motivation & consistency	Notifications, gamification, or rewards encourage users to stick to their routines.
Safe workouts	Exercise instructions, form guides, and reminders reduce the risk of injury.
Convenience	Users can work out anywhere without a personal trainer or gym.

8. Conclusion

The development of the Wise Workout App represents a strategic response to the growing demand for personalized, accessible, and intelligent fitness solutions. By combining adaptive workout planning, real-time progress tracking, and user-centric design, the app empowers individuals to achieve their health goals more effectively and safely. Through seamless integration of smart technology and evidence-based training principles, the app not only enhances user engagement and motivation but also promotes long-term fitness habits. Moving forward, continuous user feedback and data-driven improvements will be key to sustaining growth, increasing user retention, and positioning the Wise Workout App as a leading digital fitness companion in an increasingly health-conscious market.

9. References

9.1 References

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